

BFS Traversal

Step 1: Start at A

Legend

Complete

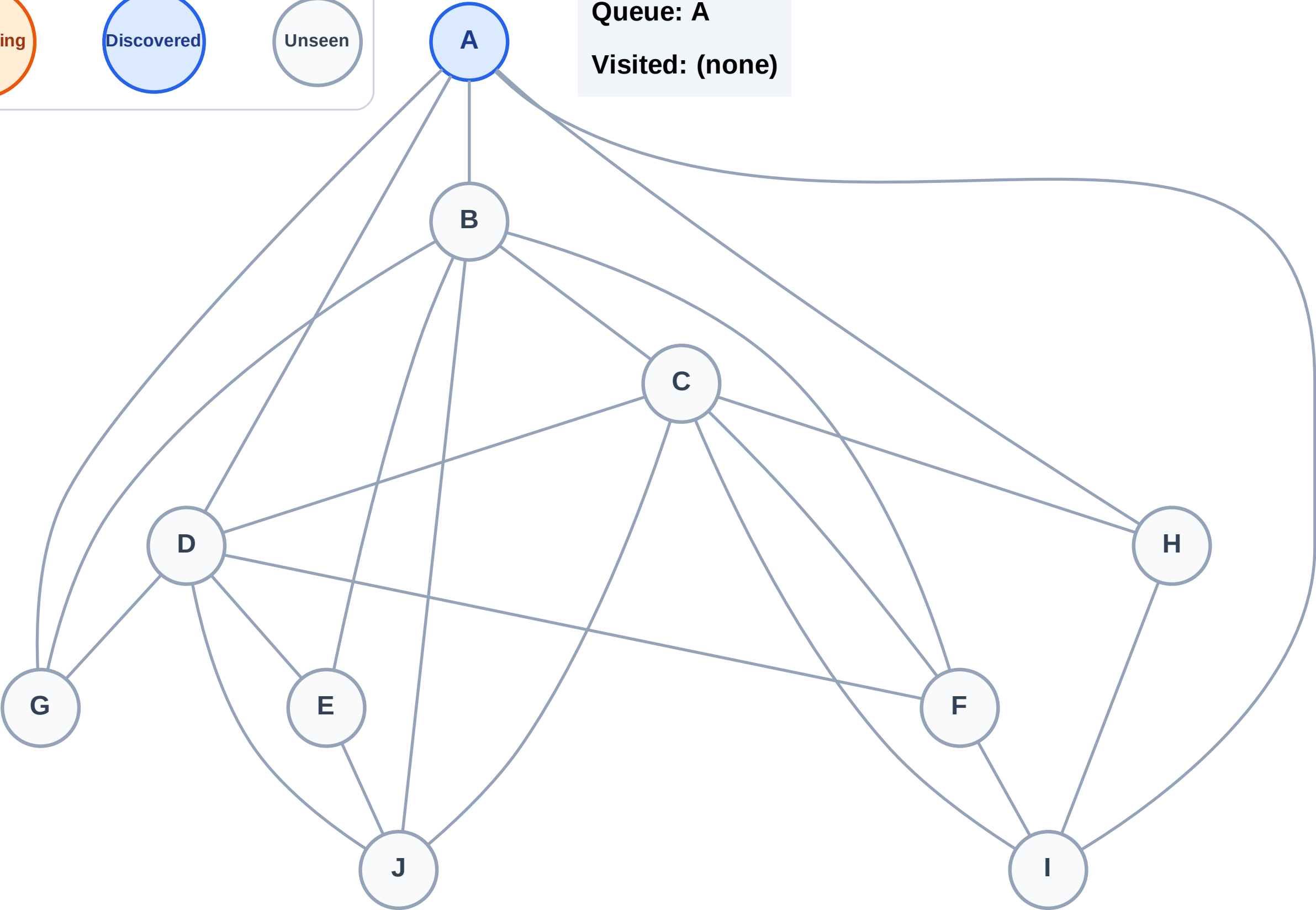
Processing

Discovered

Unseen

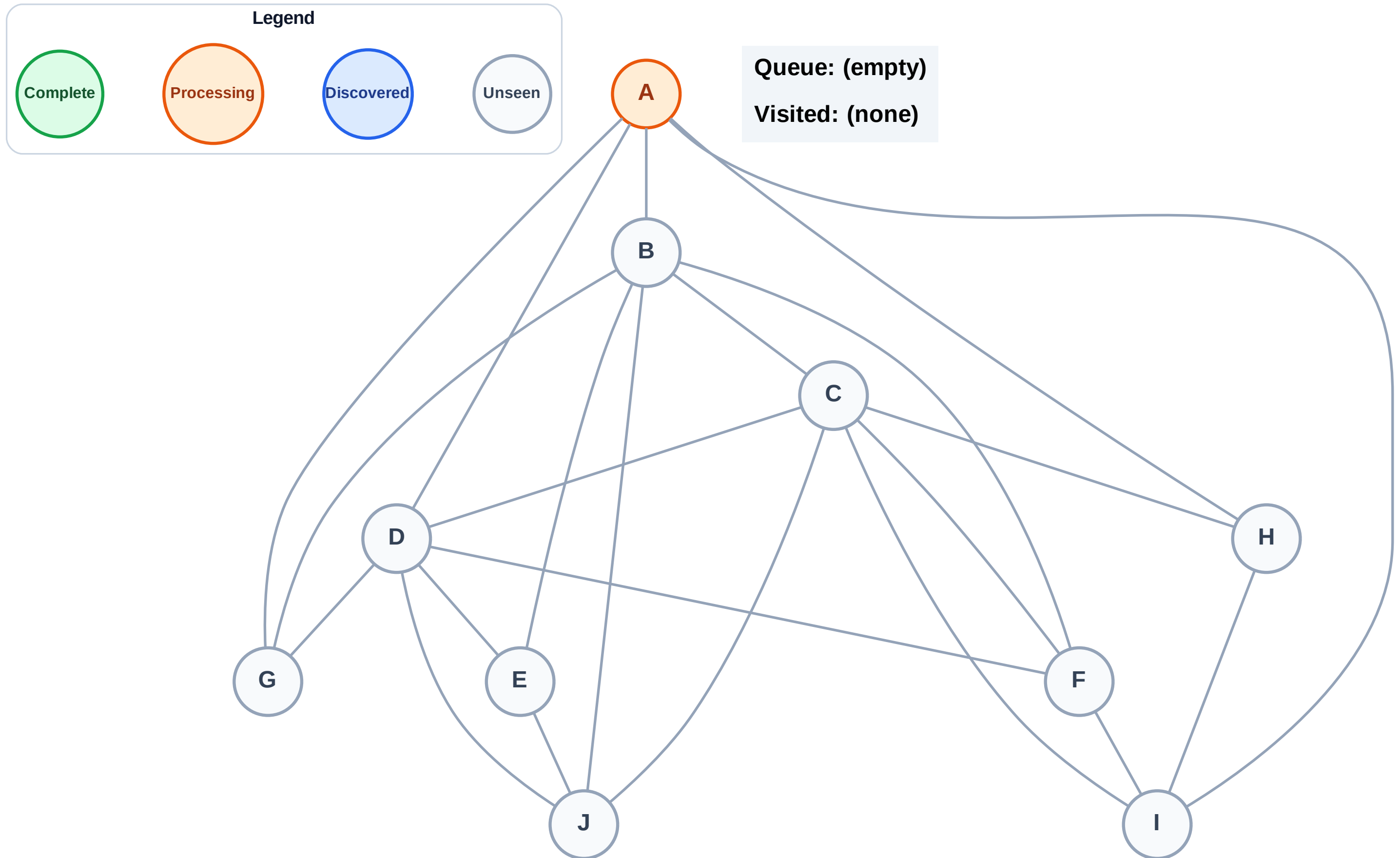
Queue: A

Visited: (none)



Step 2: Process node A

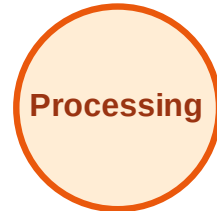
Step 2: Process node A



BFS Traversal

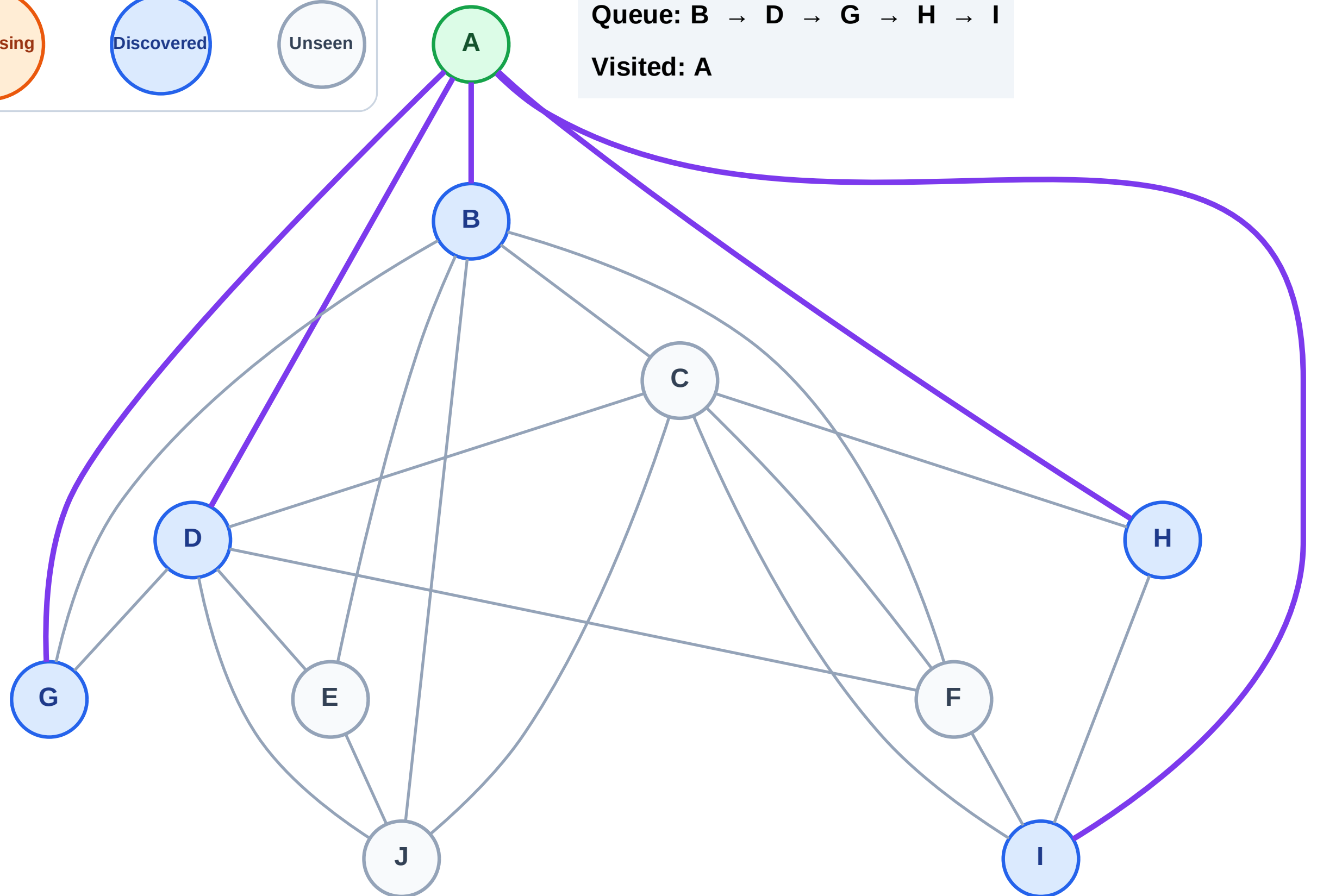
Step 3: Discover B, D, G, H, I from A

Legend



Queue: B → D → G → H → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

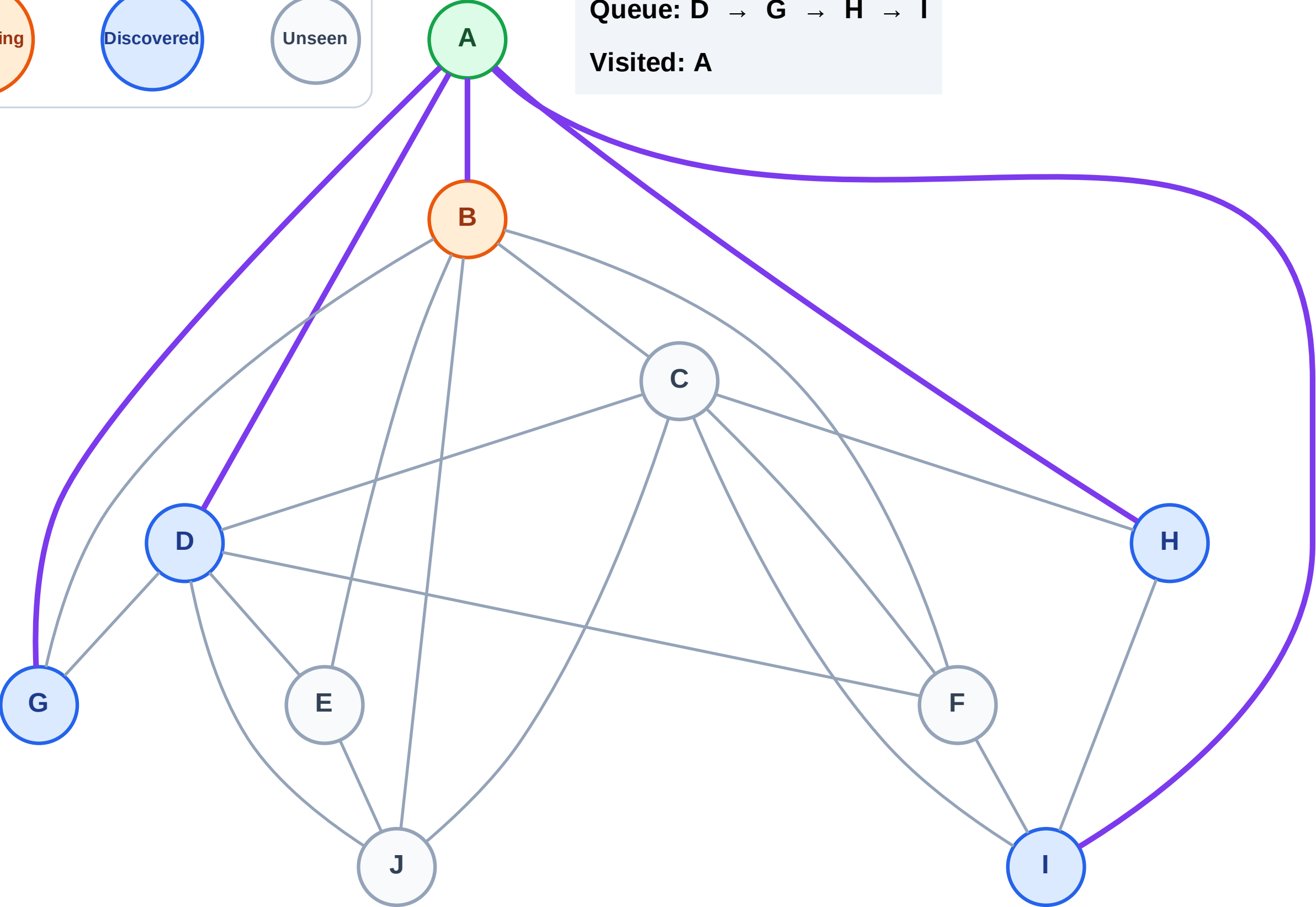
Processing

Discovered

Unseen

Queue: D → G → H → I

Visited: A



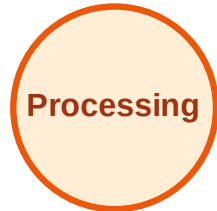
BFS Traversal

Step 5: Discover C, E, F, J from B

Legend



Complete



Processing



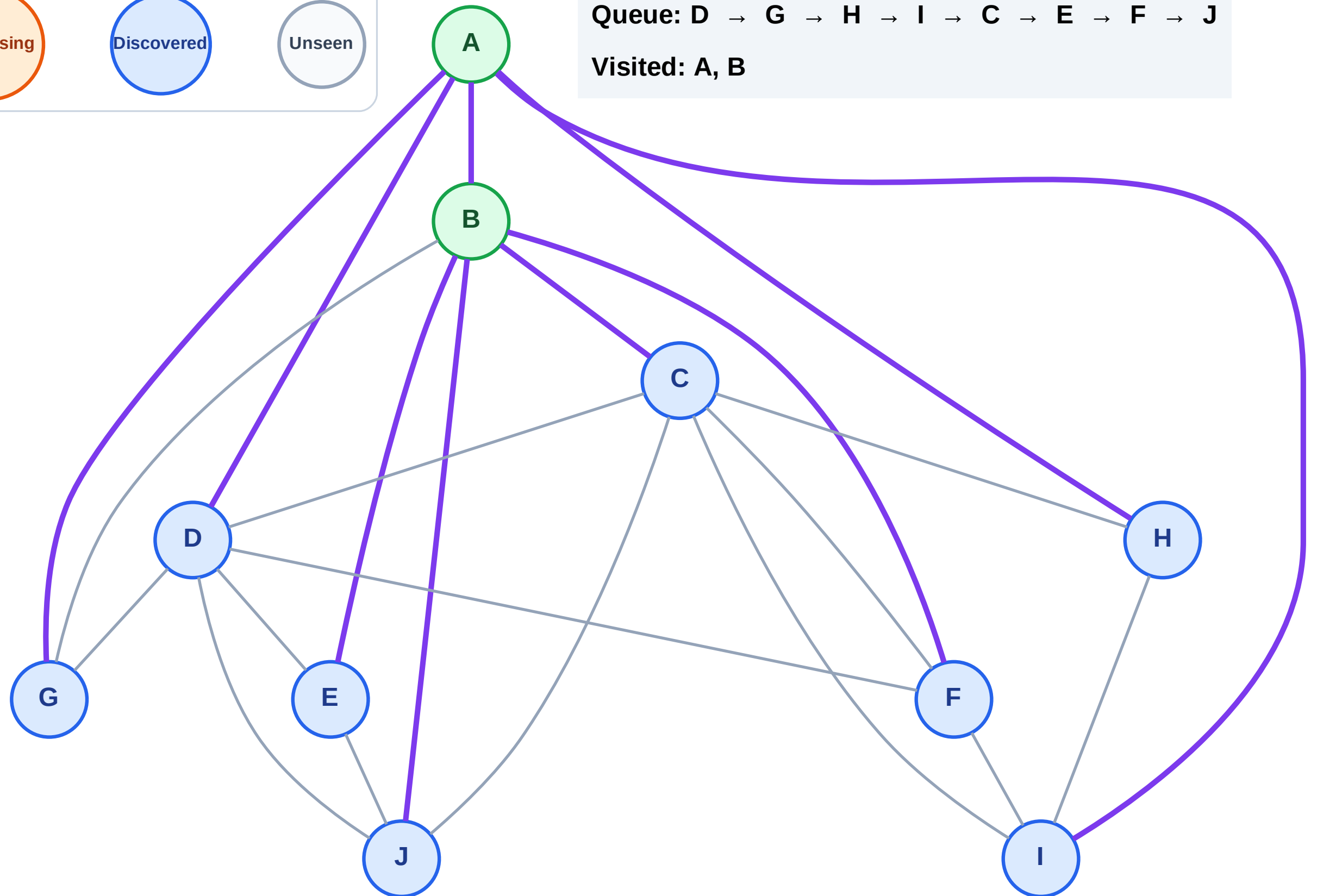
Discovered



Unseen

Queue: D → G → H → I → C → E → F → J

Visited: A, B



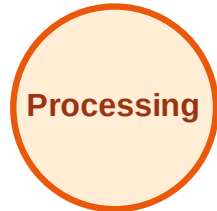
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



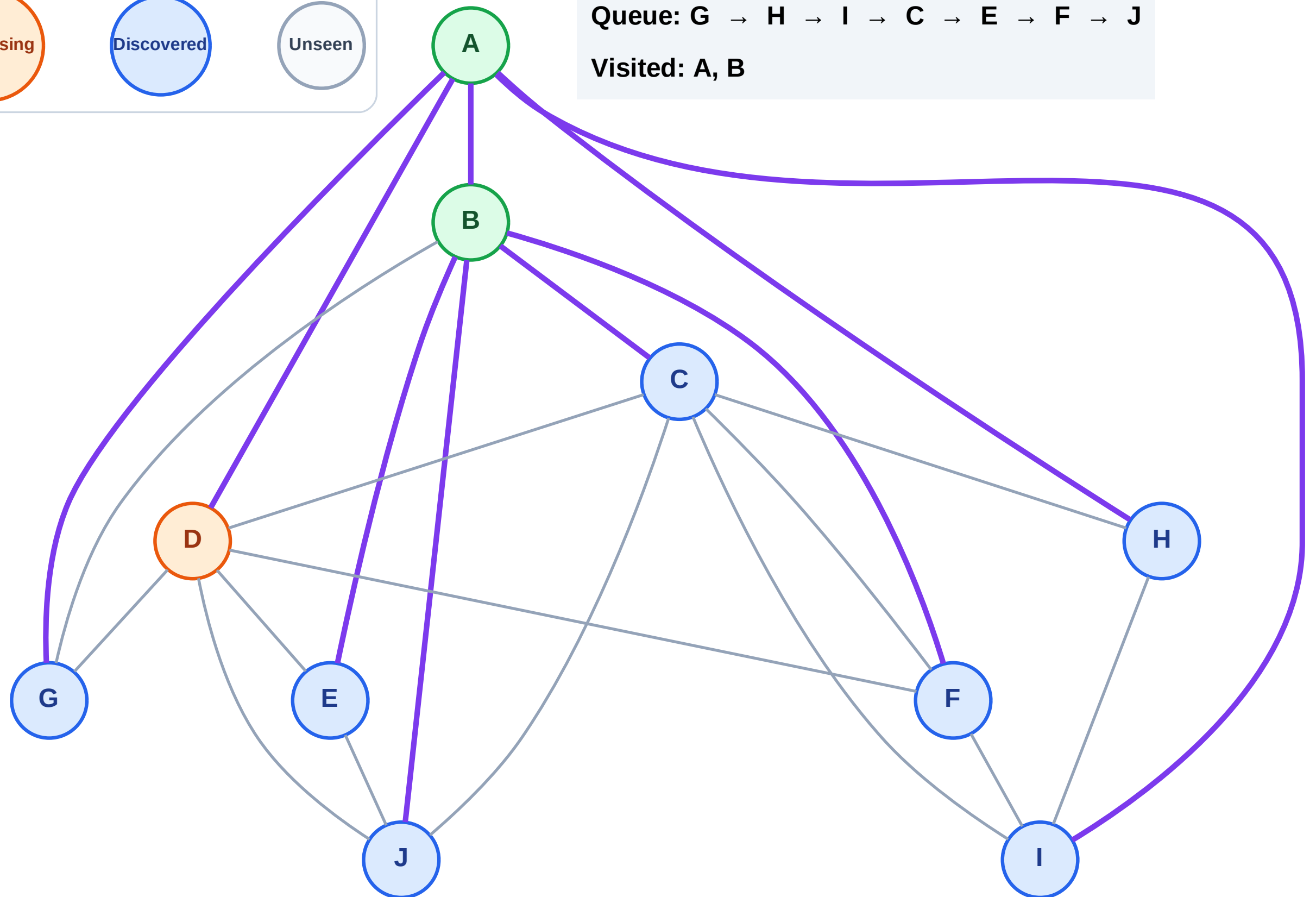
Discovered



Unseen

Queue: G → H → I → C → E → F → J

Visited: A, B



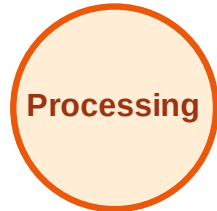
BFS Traversal

Step 7: No new neighbors from D

Legend



Complete



Processing



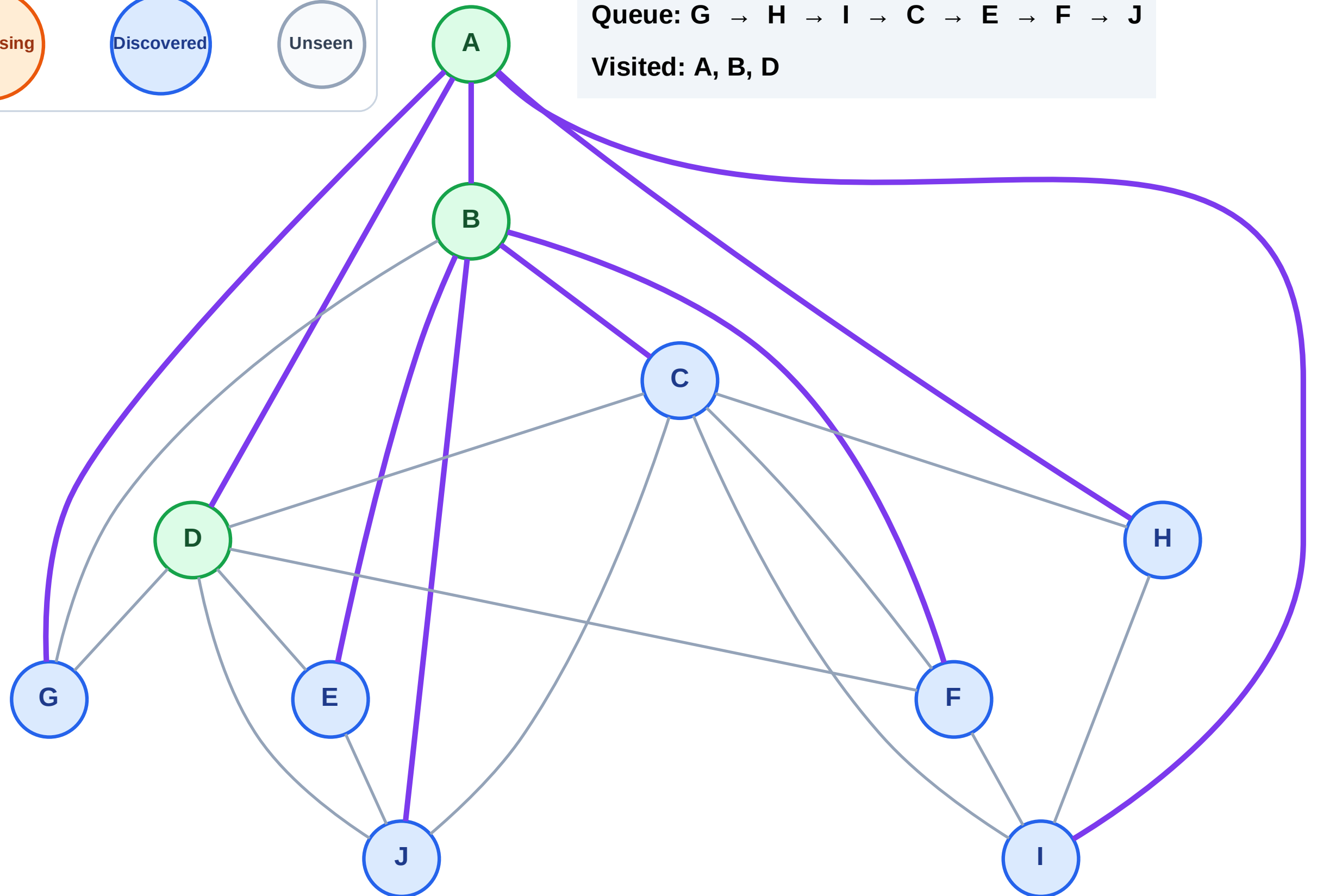
Discovered



Unseen

Queue: G → H → I → C → E → F → J

Visited: A, B, D



BFS Traversal

Step 8: Process node G

Legend

Complete

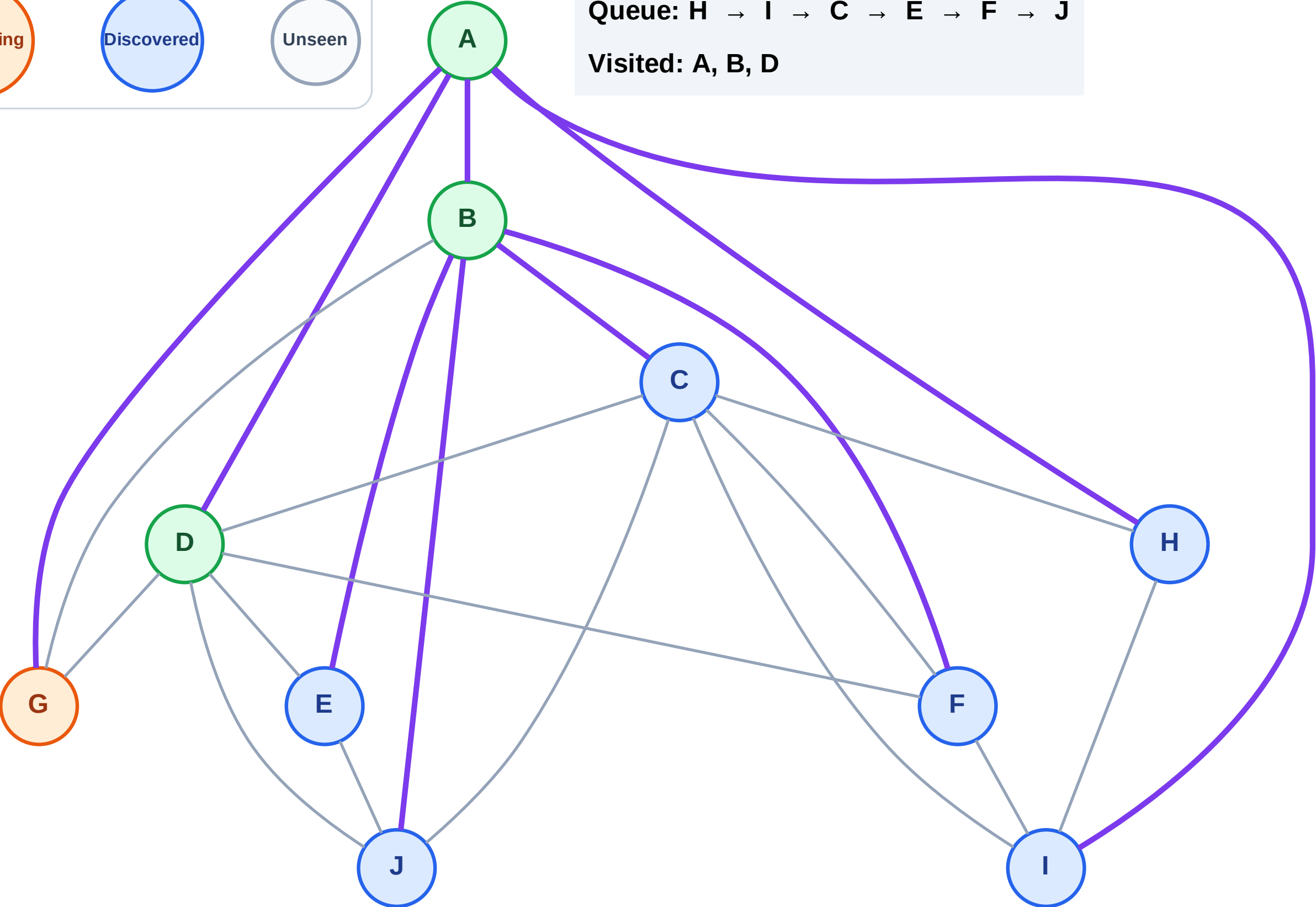
Processing

Discovered

Unseen

Queue: H → I → C → E → F → J

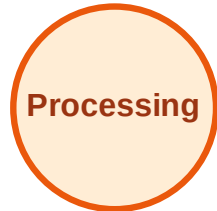
Visited: A, B, D



BFS Traversal

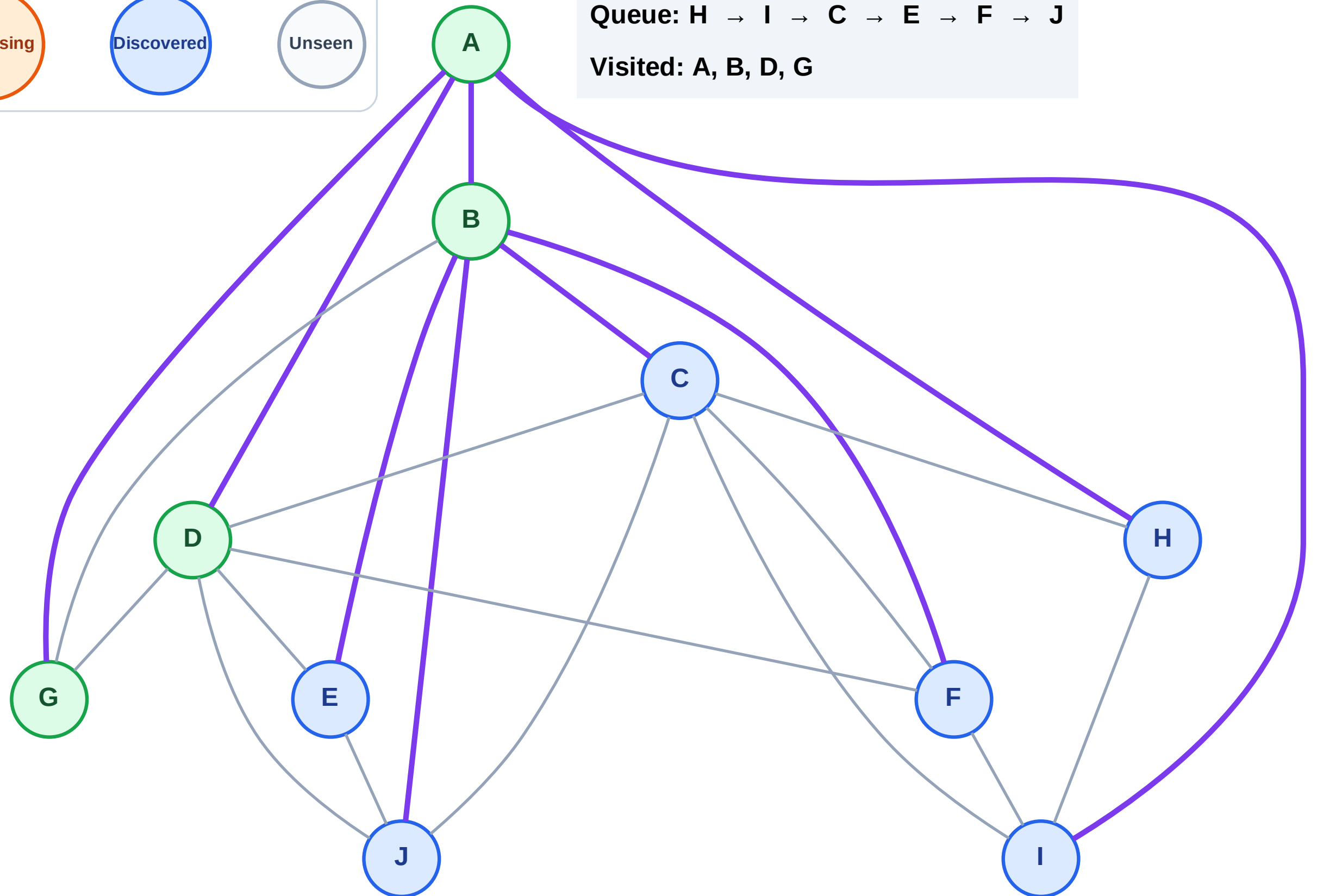
Step 9: No new neighbors from G

Legend



Queue: H → I → C → E → F → J

Visited: A, B, D, G



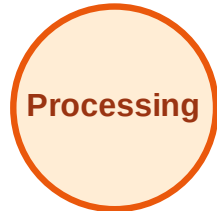
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



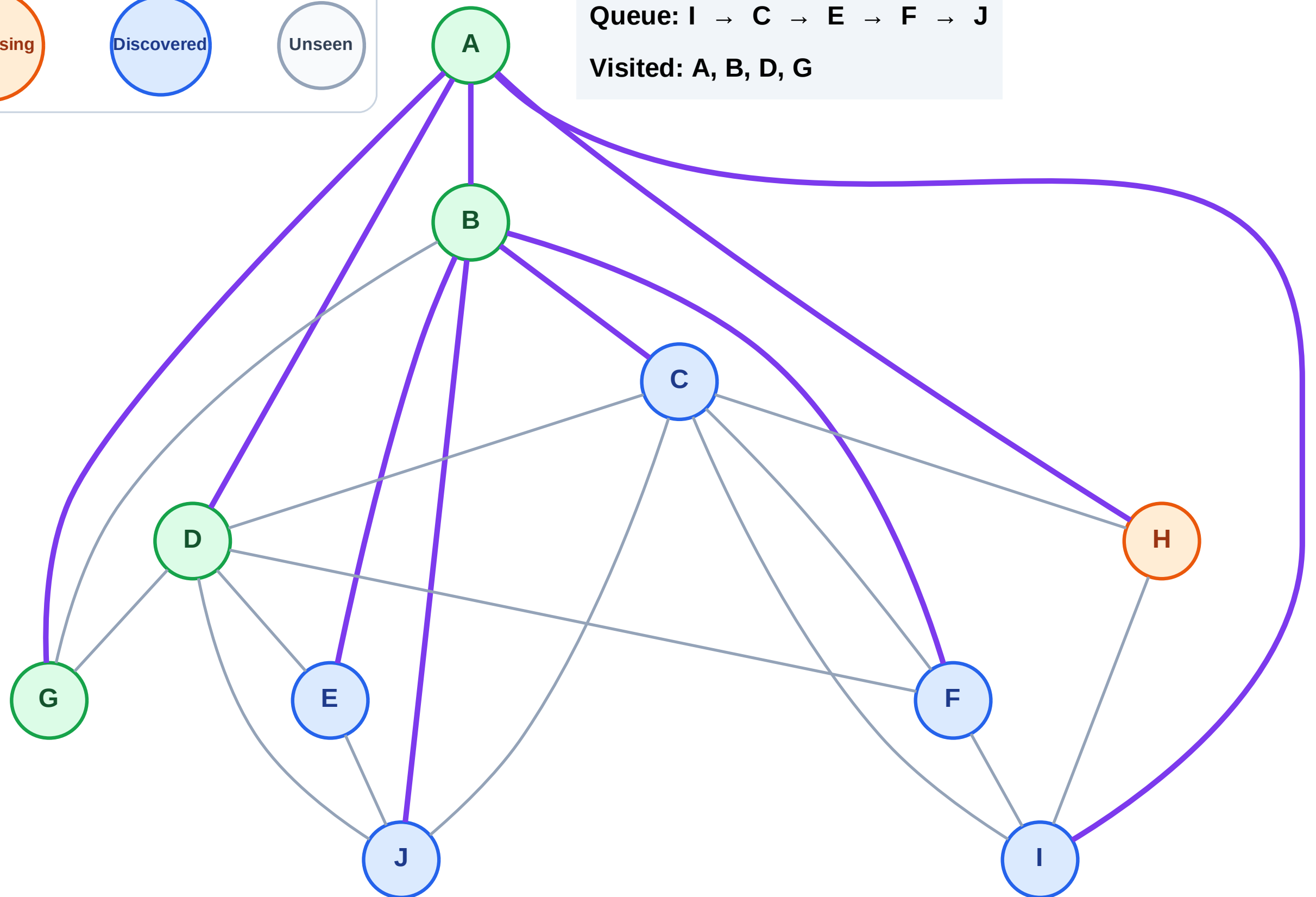
Discovered



Unseen

Queue: I → C → E → F → J

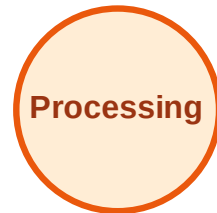
Visited: A, B, D, G



BFS Traversal

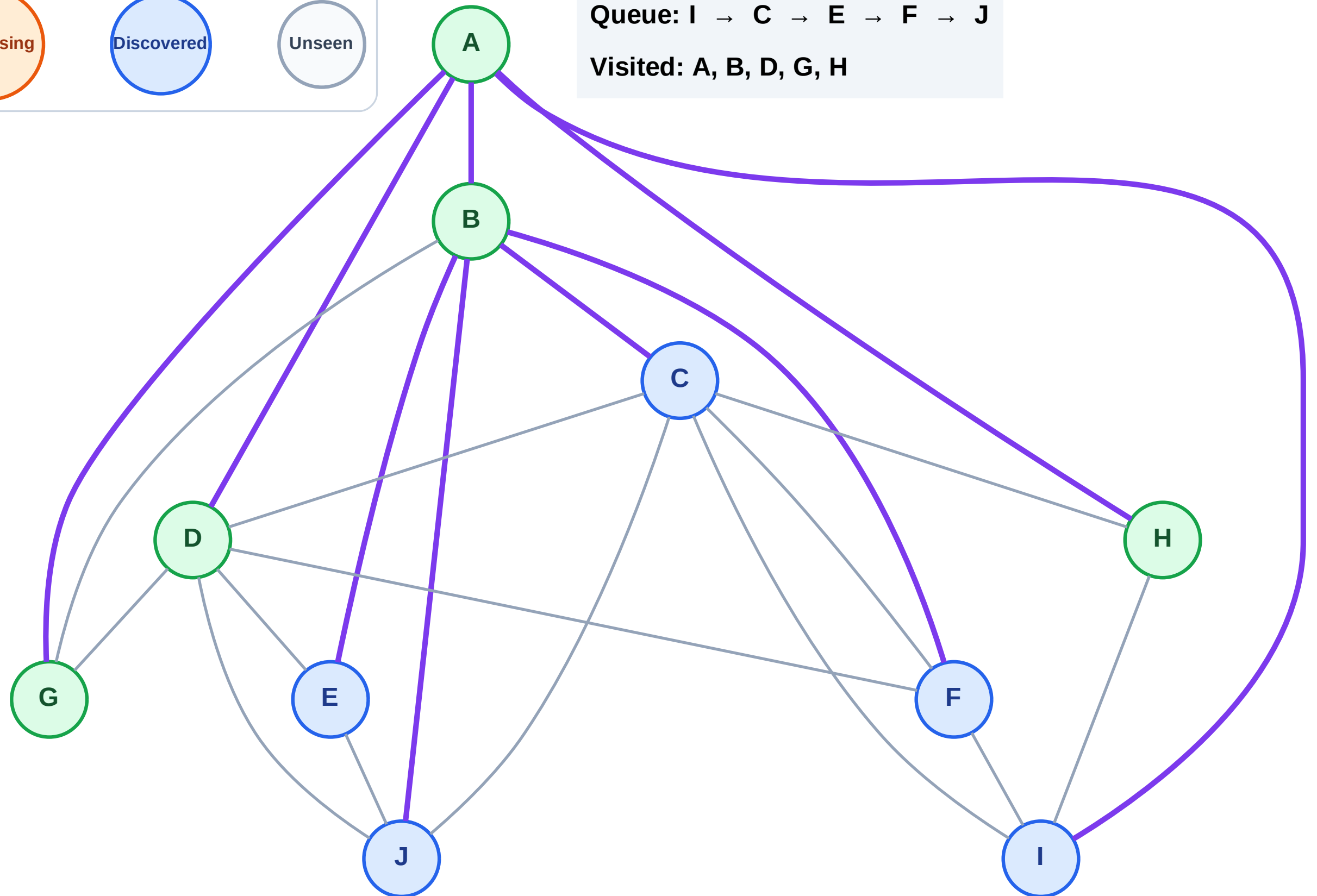
Step 11: No new neighbors from H

Legend



Queue: I → C → E → F → J

Visited: A, B, D, G, H



BFS Traversal

Step 12: Process node I

Legend

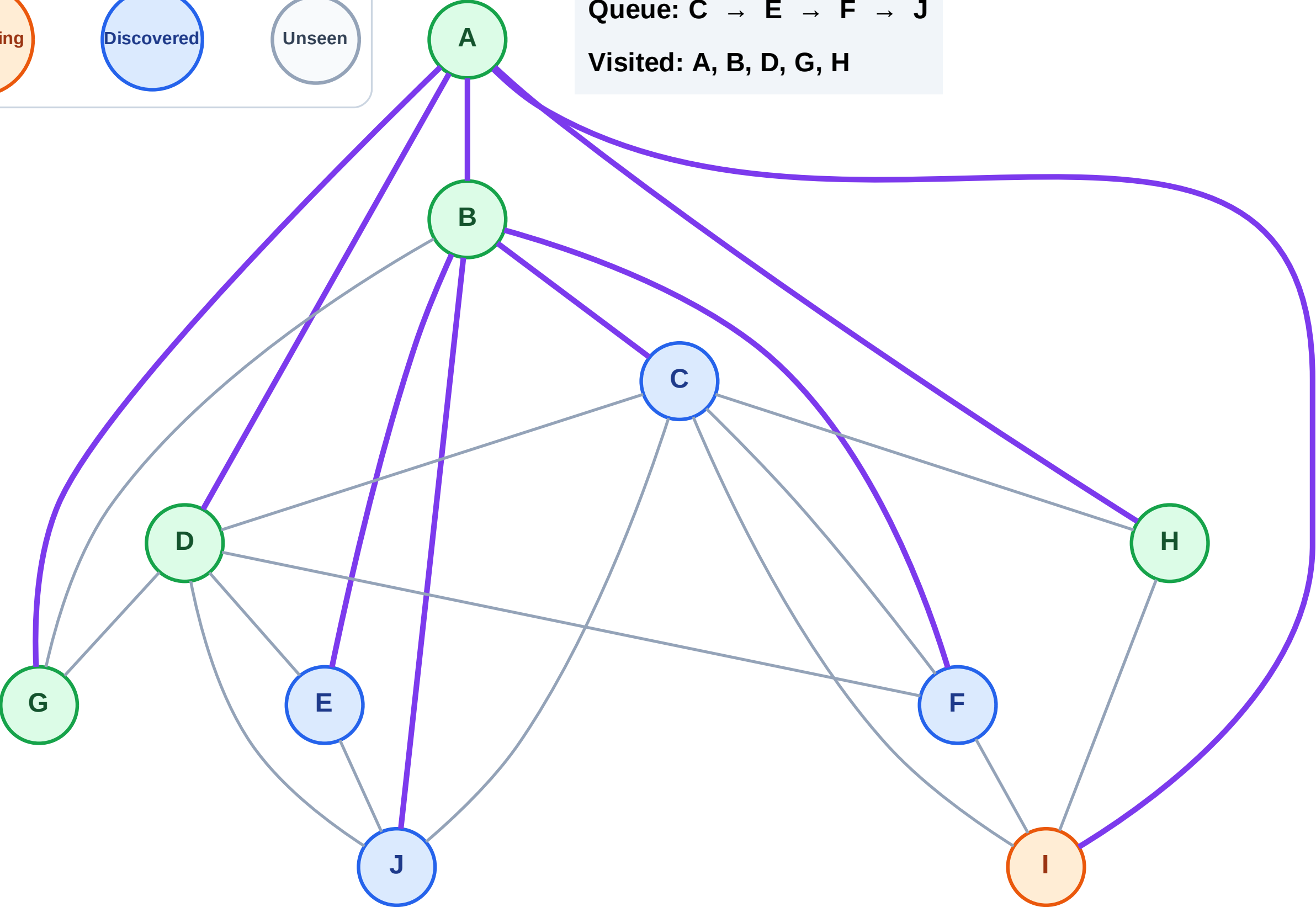
Complete

Processing

Discovered

Unseen

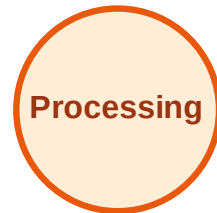
Queue: C → E → F → J
Visited: A, B, D, G, H



BFS Traversal

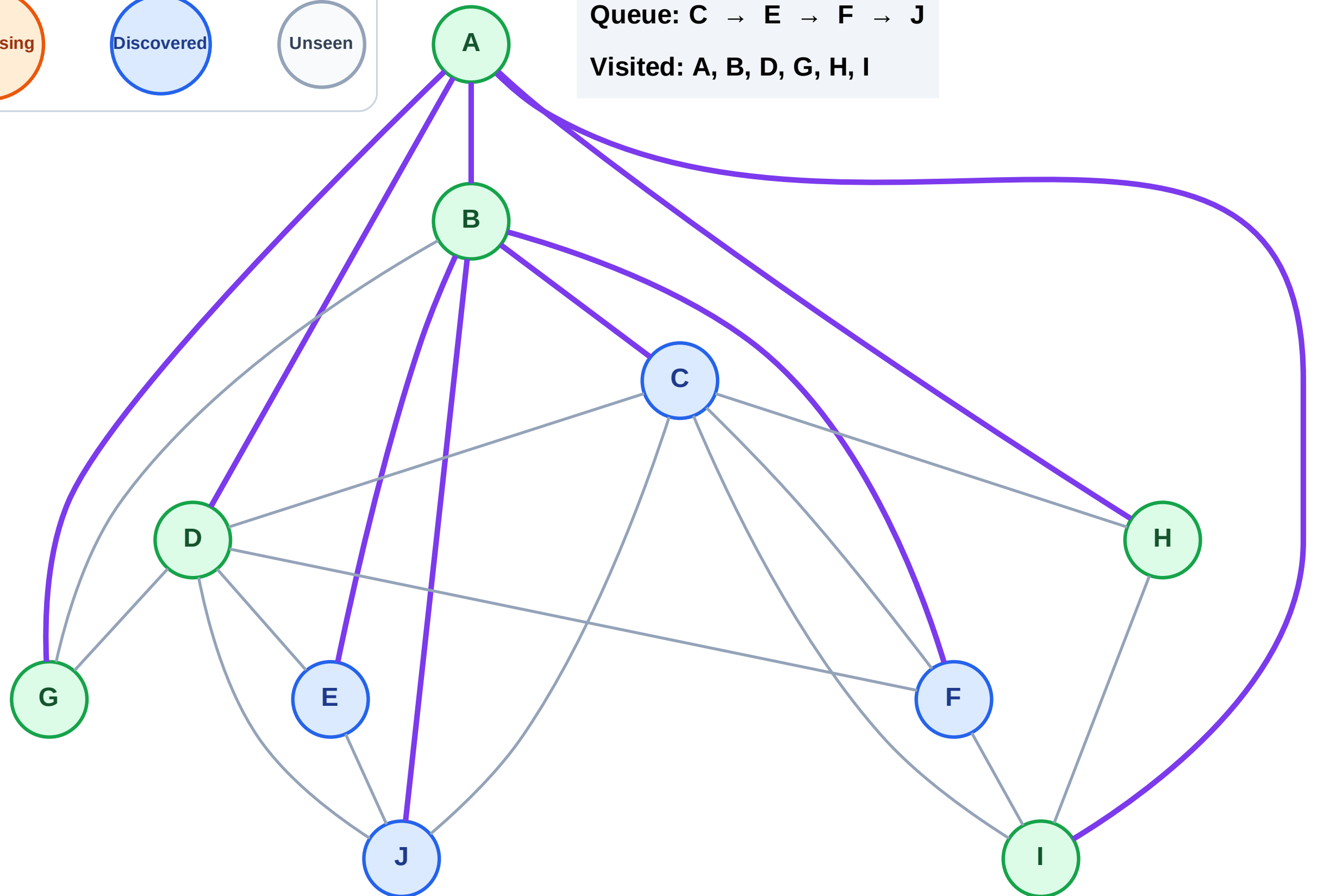
Step 13: No new neighbors from I

Legend



Queue: C → E → F → J

Visited: A, B, D, G, H, I



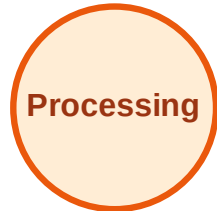
BFS Traversal

Step 14: Process node C

Legend



Complete



Processing



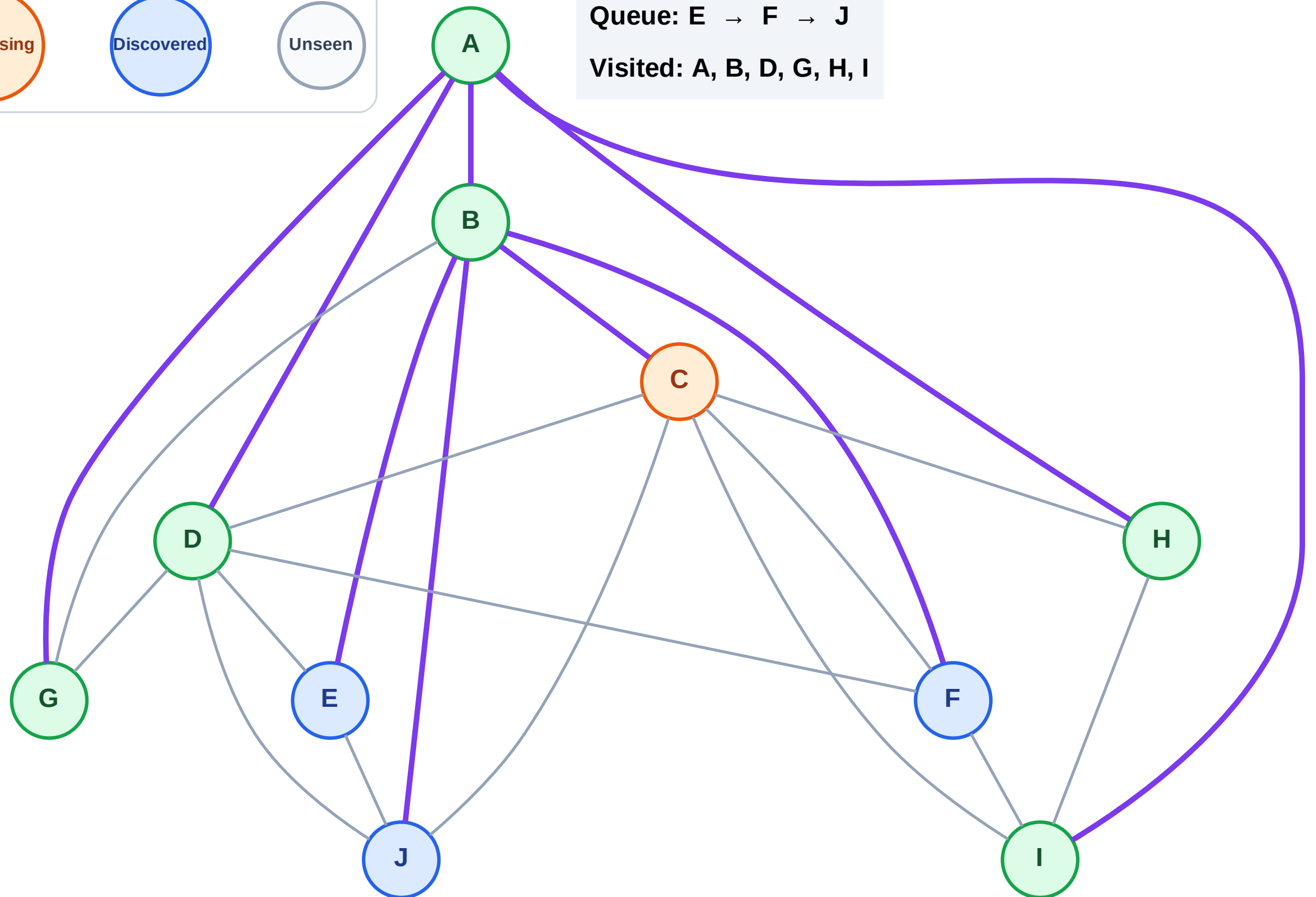
Discovered



Unseen

Queue: E → F → J

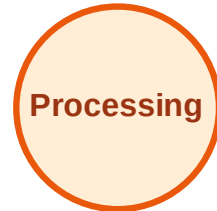
Visited: A, B, D, G, H, I



BFS Traversal

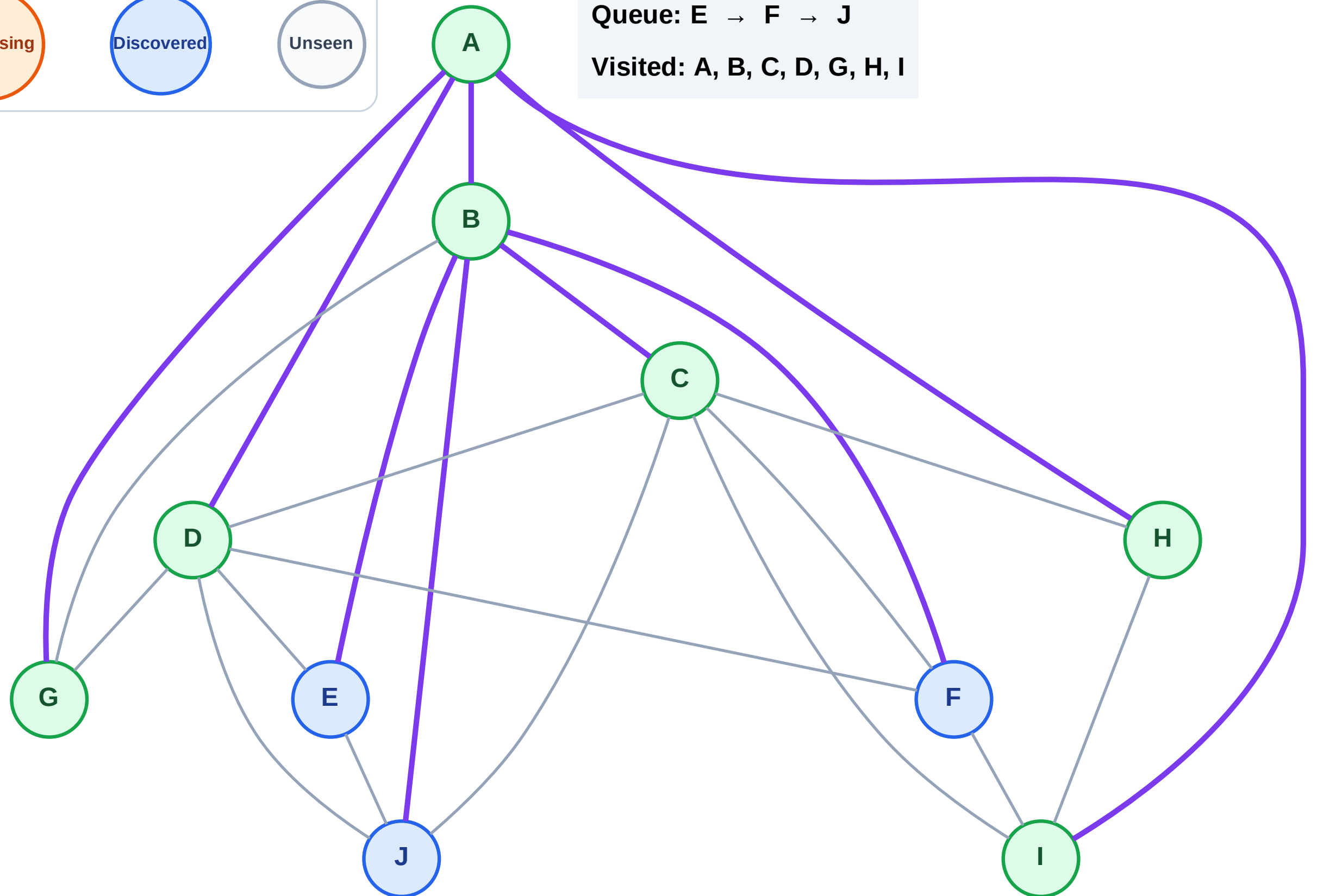
Step 15: No new neighbors from C

Legend



Queue: E → F → J

Visited: A, B, C, D, G, H, I



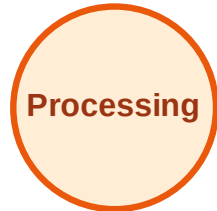
BFS Traversal

Step 16: Process node E

Legend



Complete



Processing



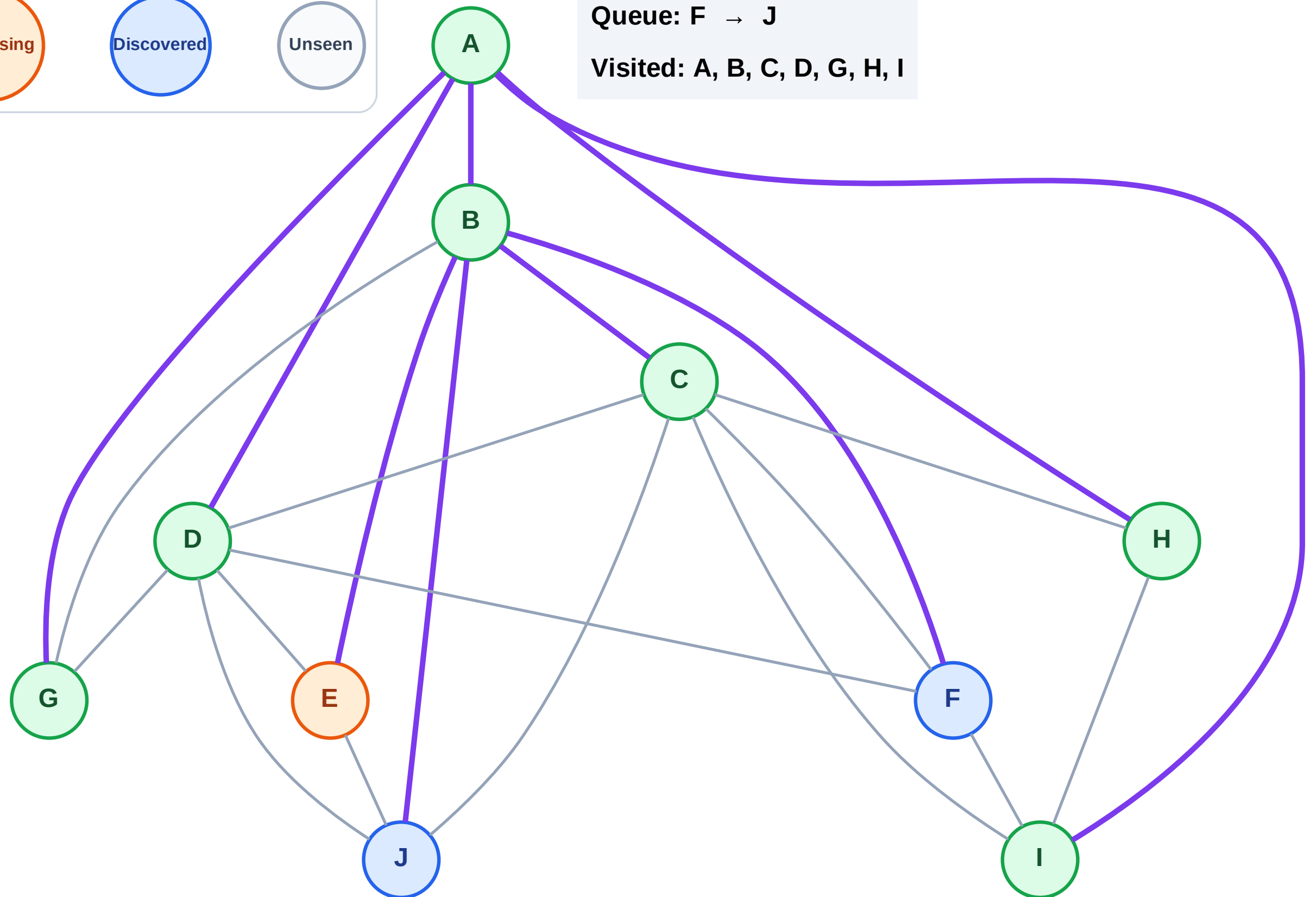
Discovered



Unseen

Queue: F → J

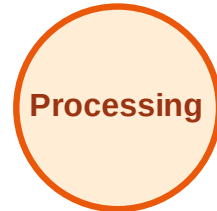
Visited: A, B, C, D, G, H, I



BFS Traversal

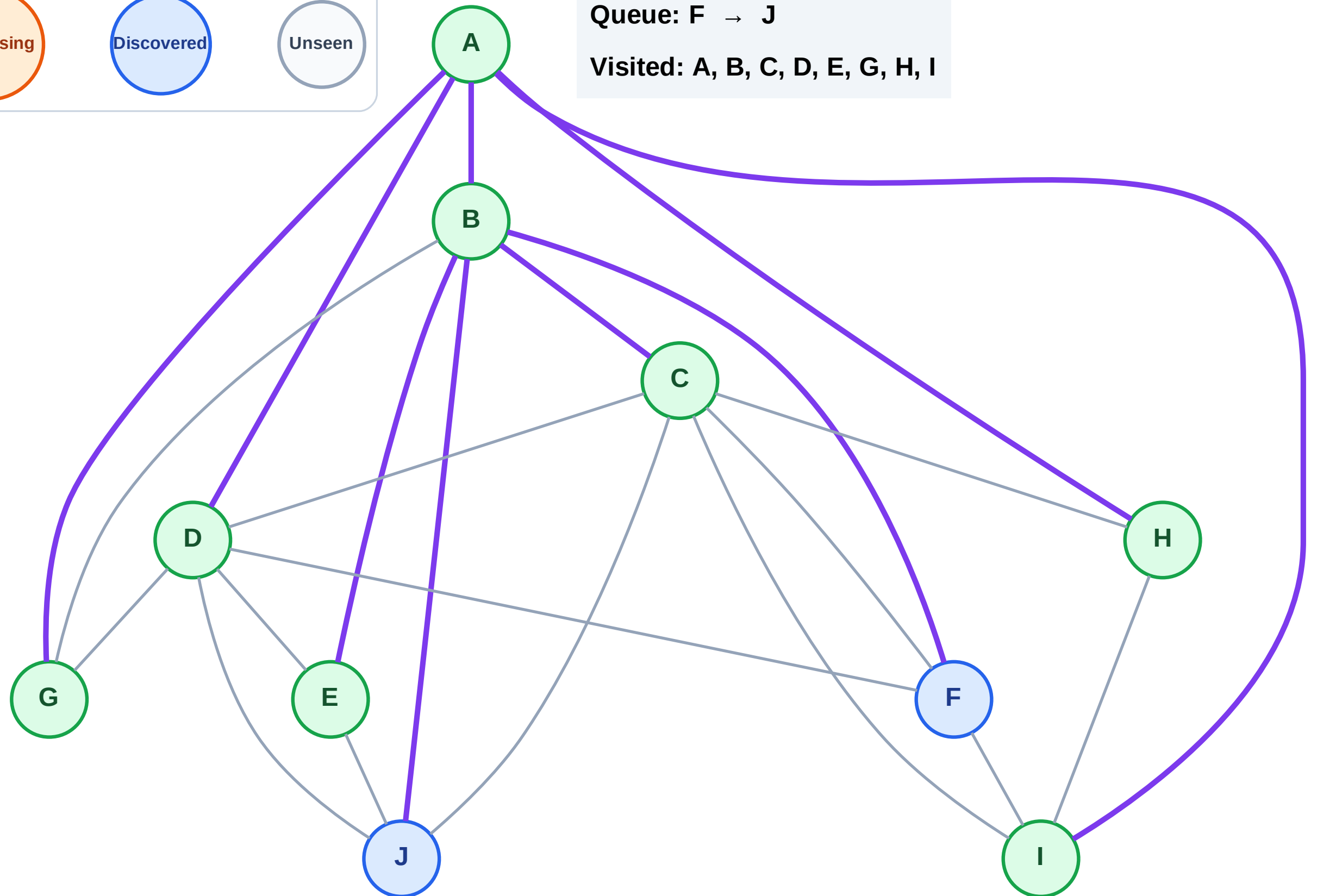
Step 17: No new neighbors from E

Legend



Queue: F → J

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node F

Legend

Complete

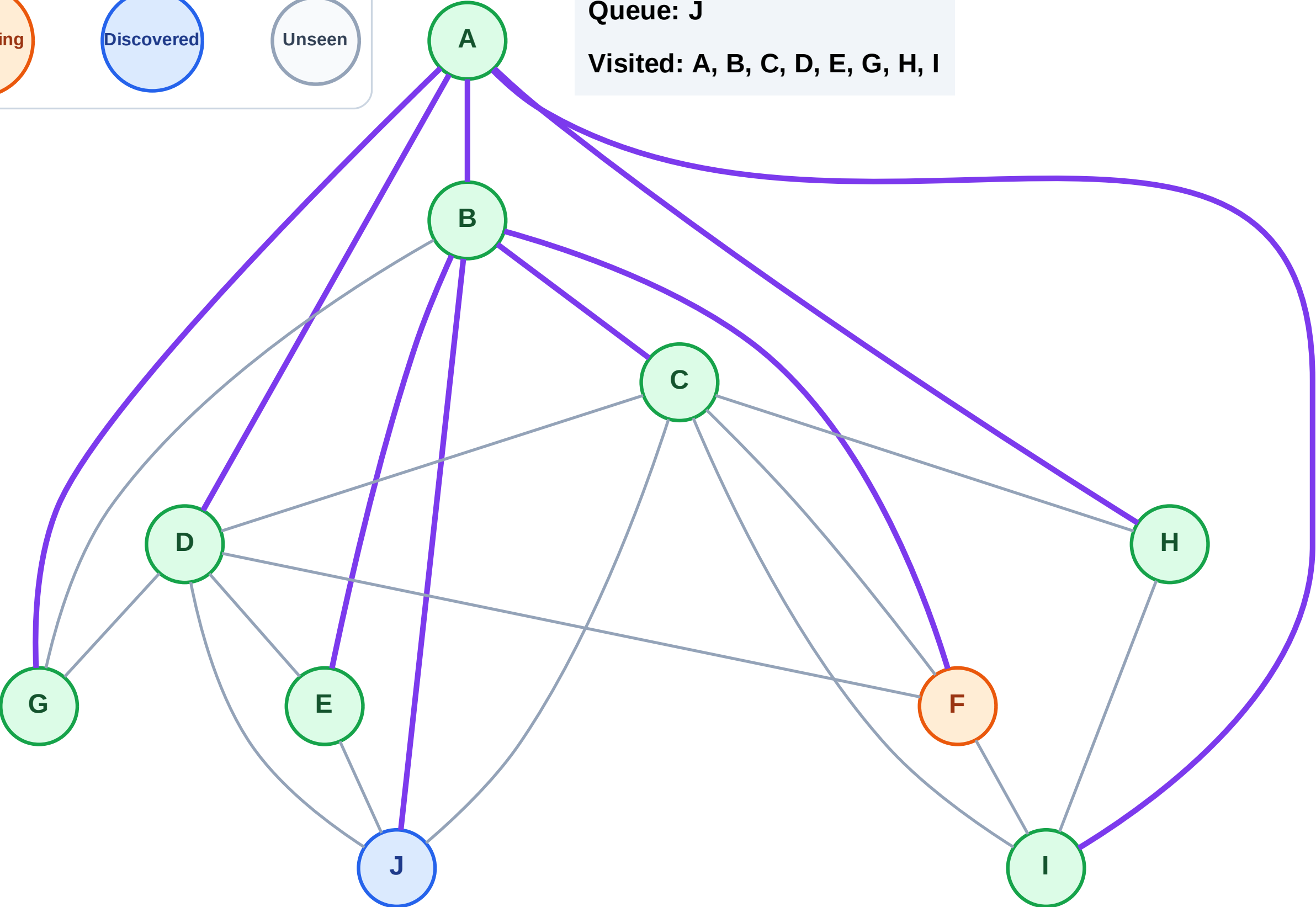
Processing

Discovered

Unseen

Queue: J

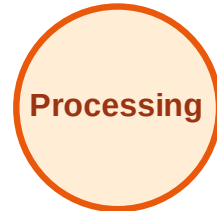
Visited: A, B, C, D, E, G, H, I



BFS Traversal

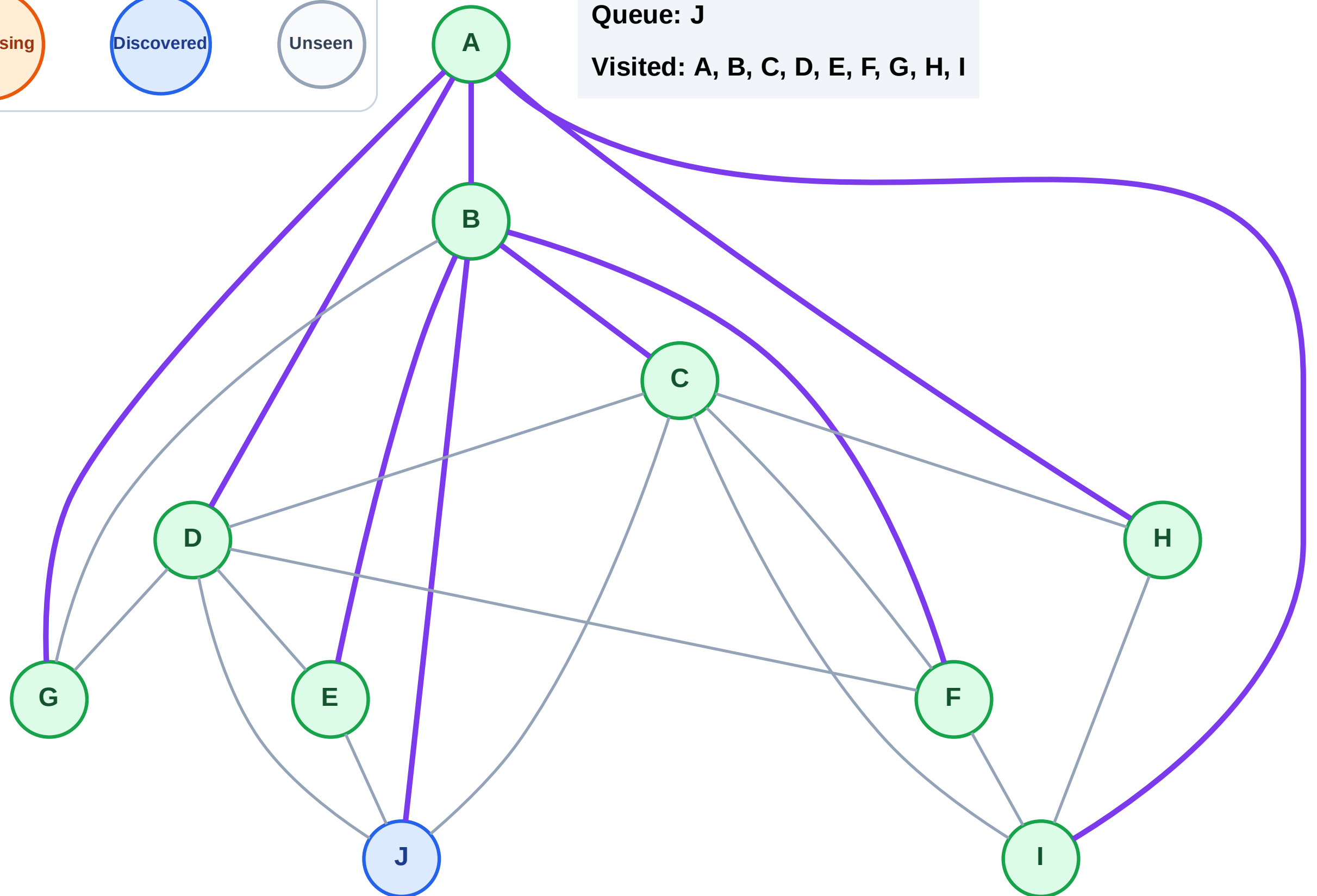
Step 19: No new neighbors from F

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

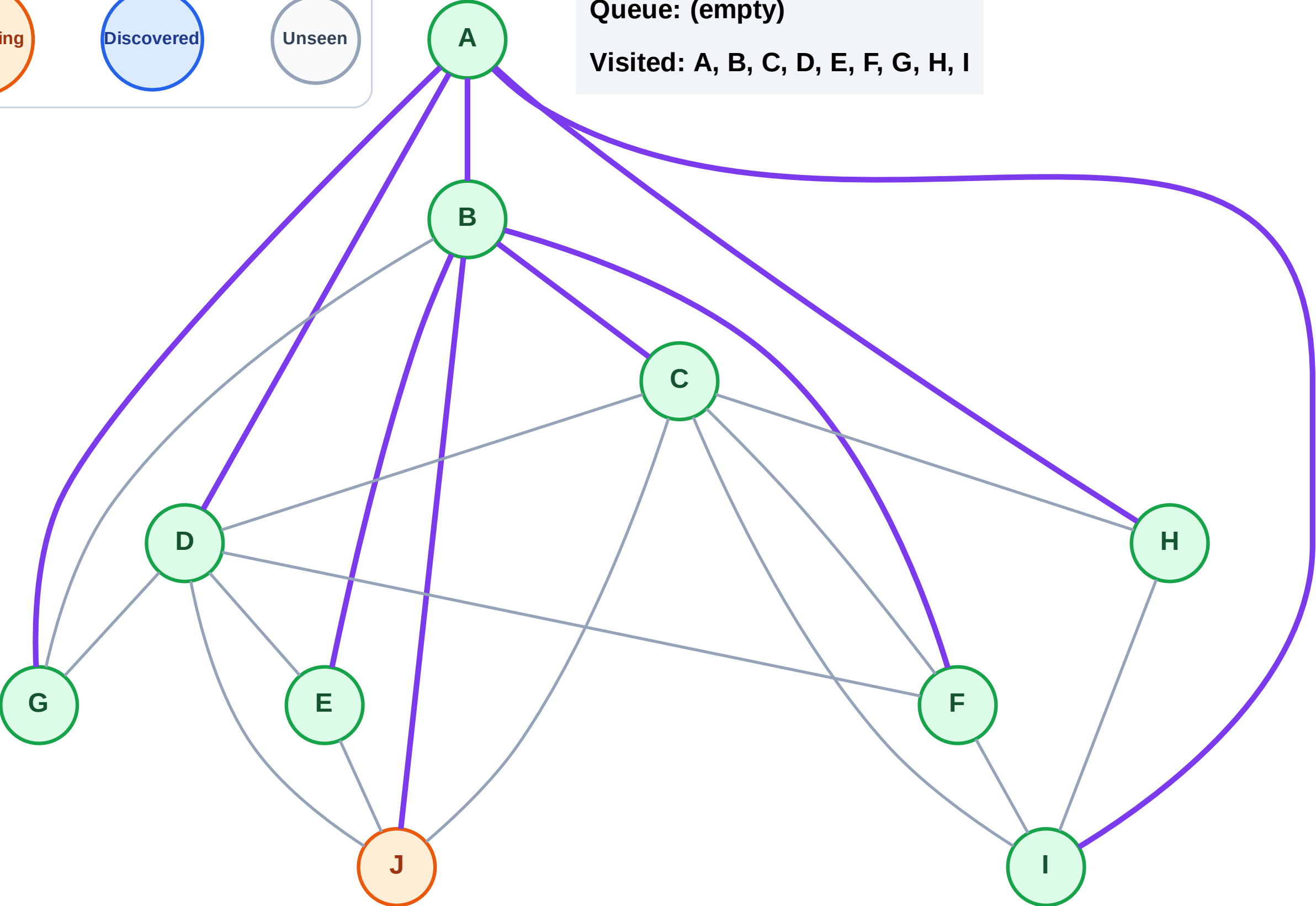
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



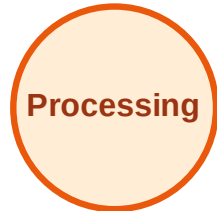
BFS Traversal

Step 21: No new neighbors from J

Legend



Complete



Processing



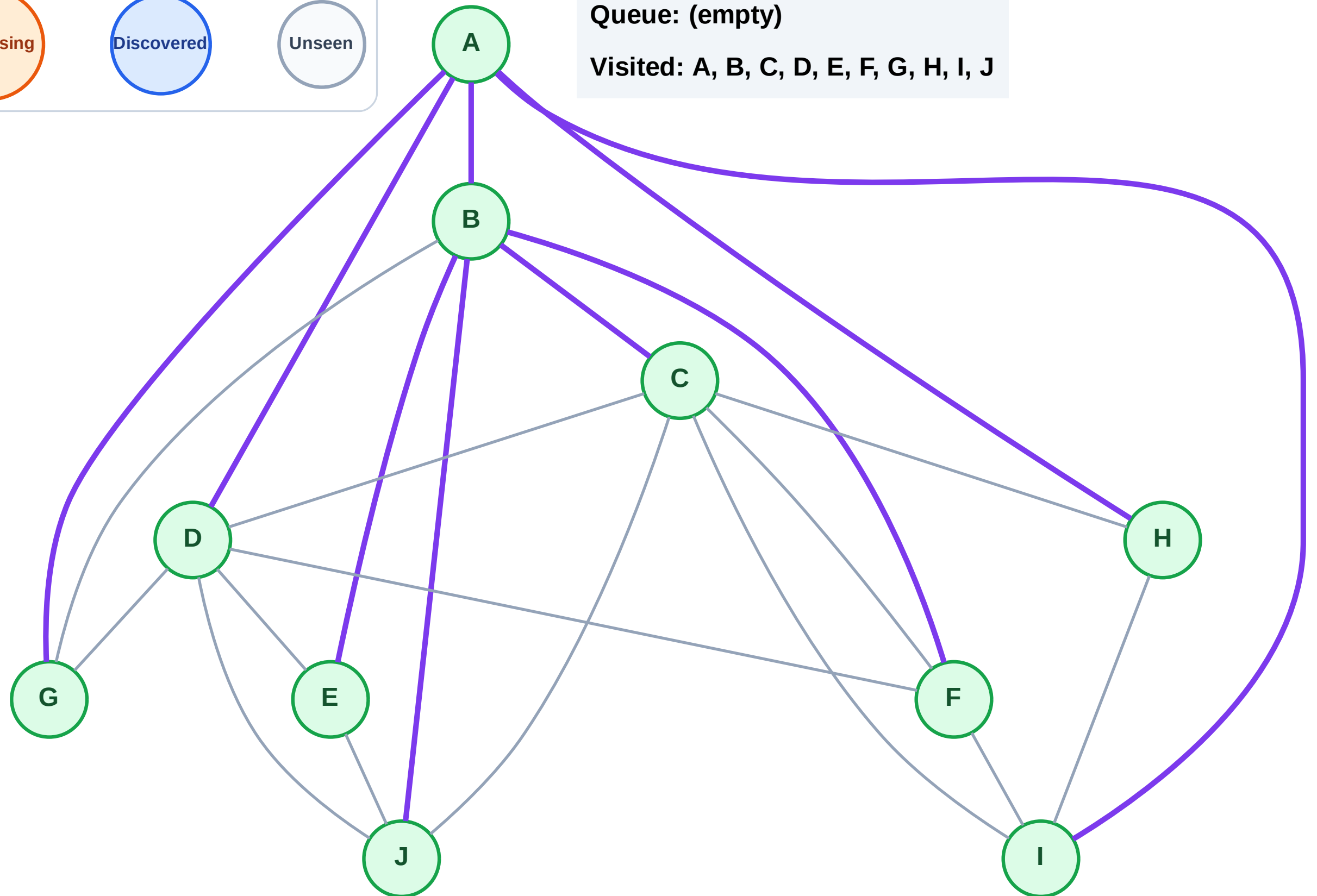
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

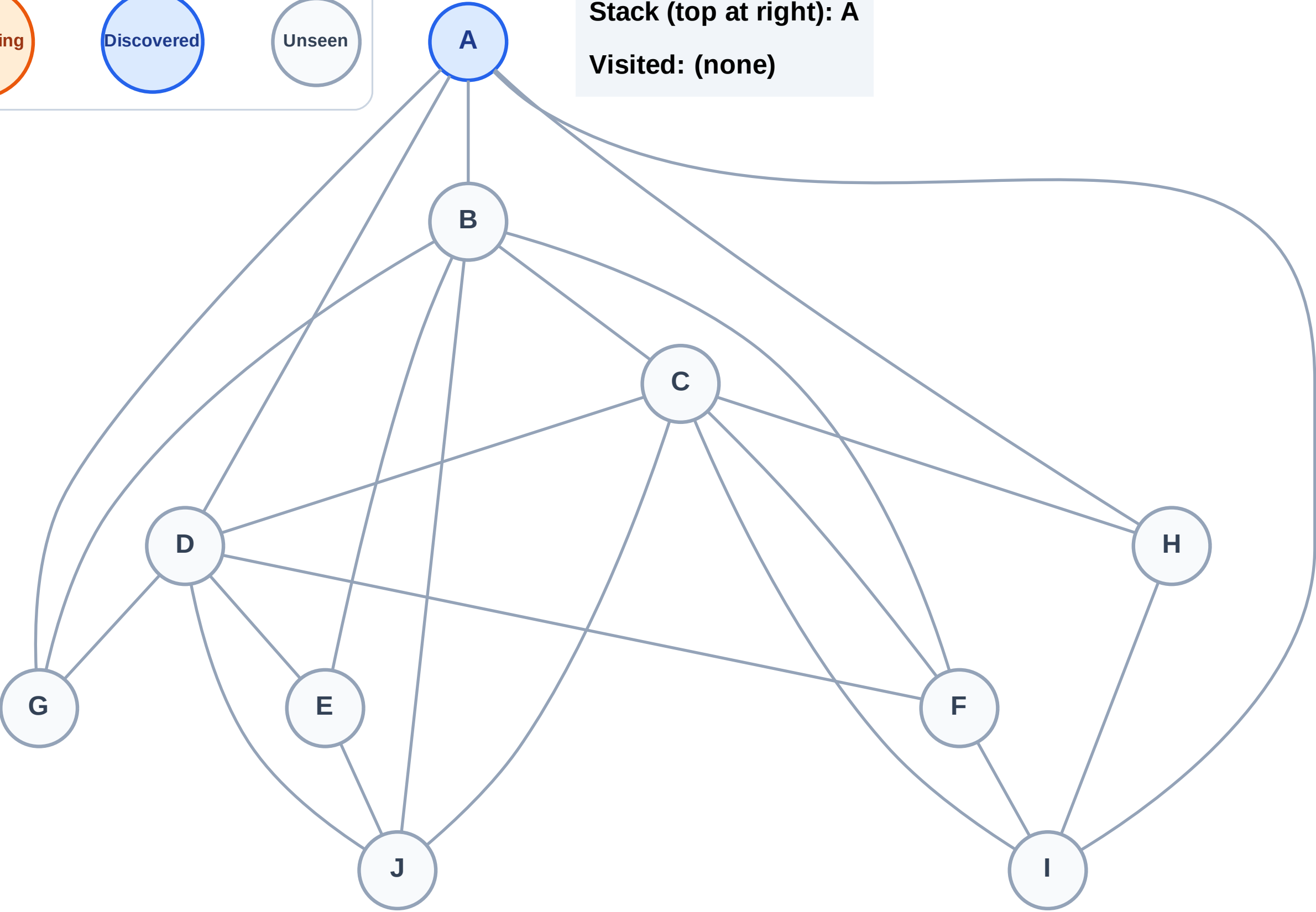
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

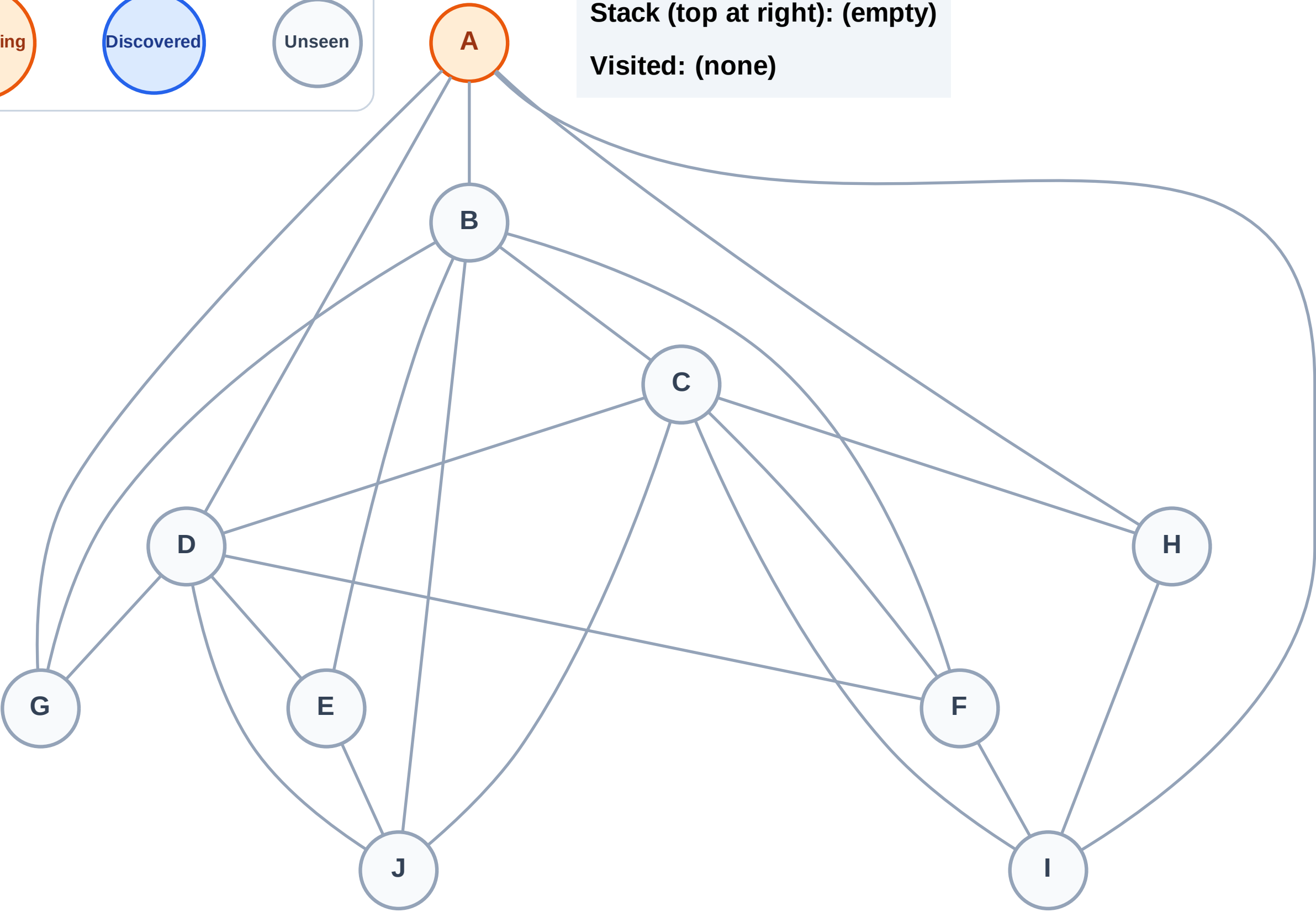
Complete

Processing

Discovered

Unseen

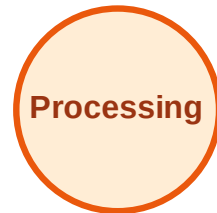
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

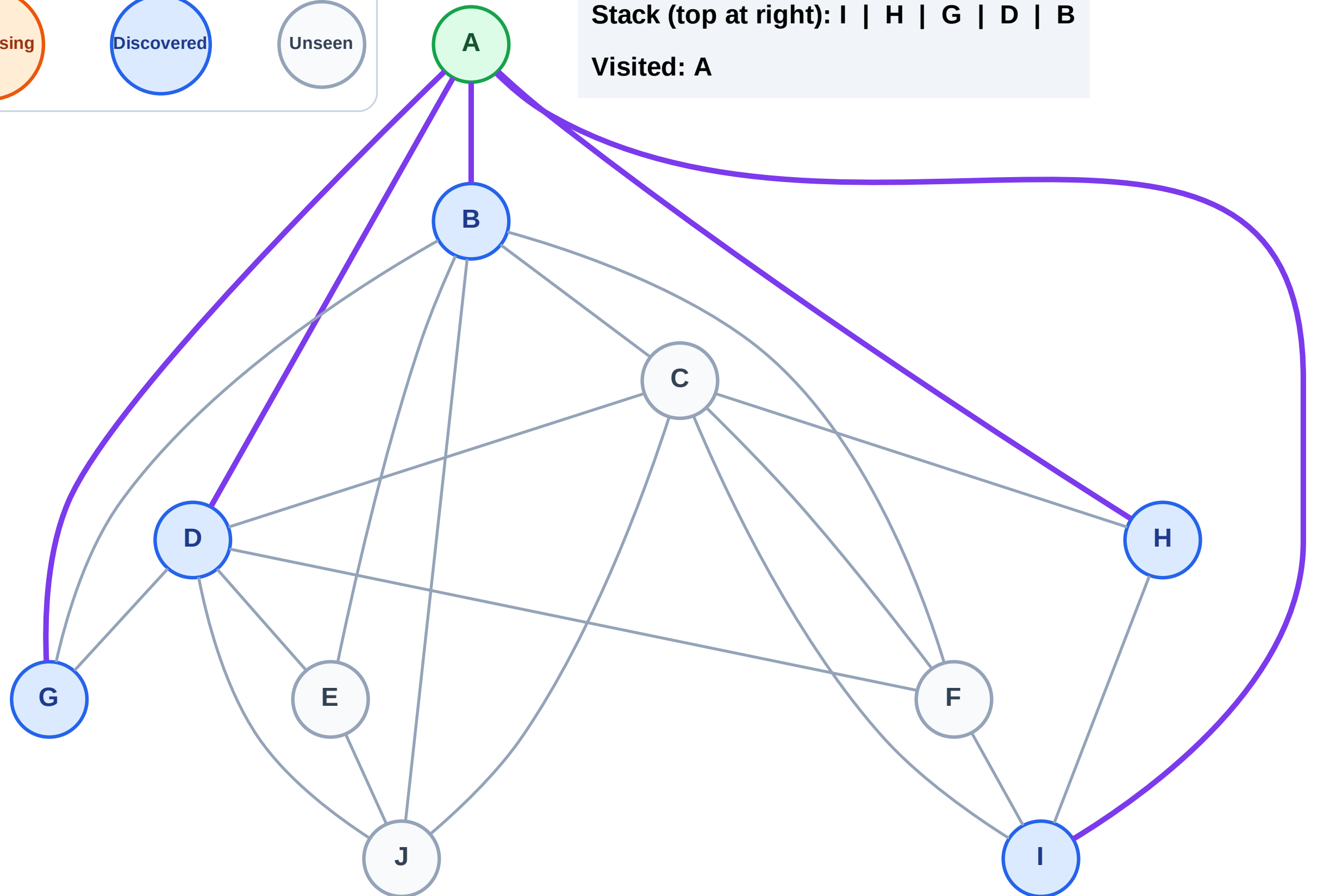
Step 3: Discover I, H, G, D, B from A

Legend



Stack (top at right): I | H | G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

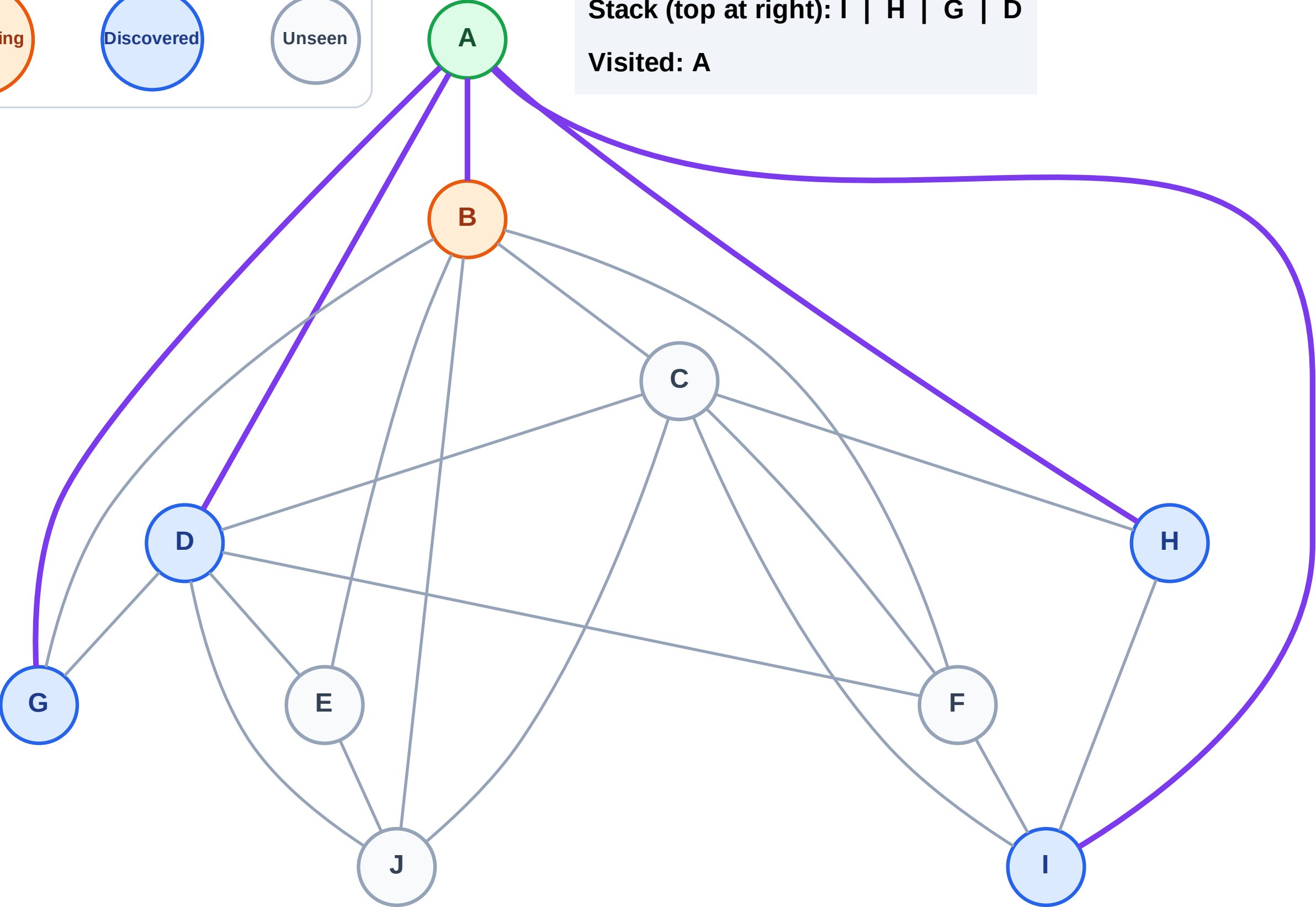
Complete

Processing

Discovered

Unseen

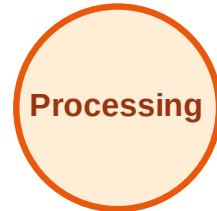
Stack (top at right): I | H | G | D
Visited: A



DFS Traversal

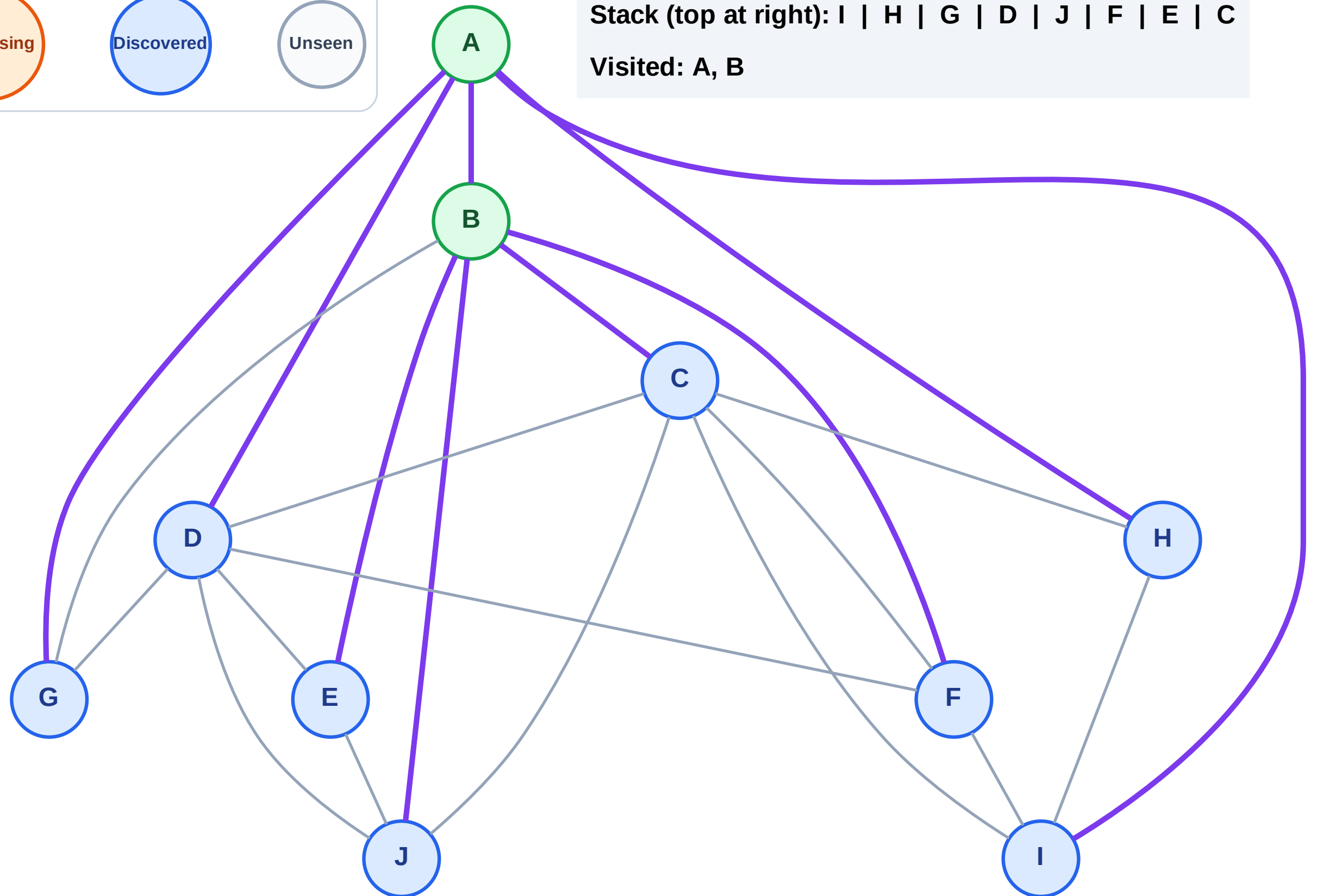
Step 5: Discover J, F, E, C from B

Legend



Stack (top at right): I | H | G | D | J | F | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

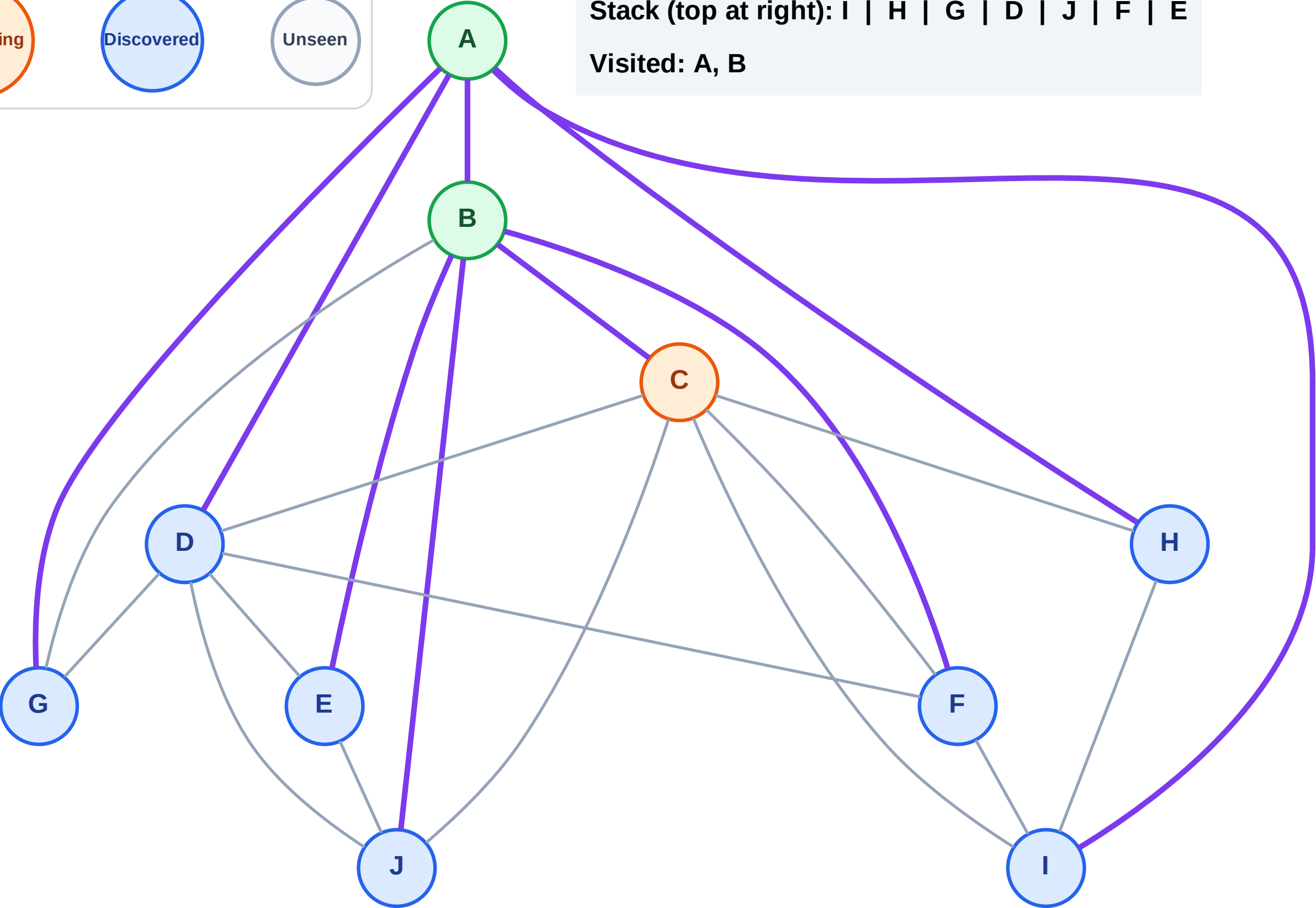
Processing

Discovered

Unseen

Stack (top at right): I | H | G | D | J | F | E

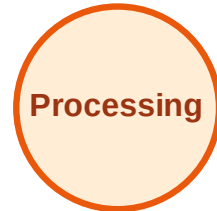
Visited: A, B



DFS Traversal

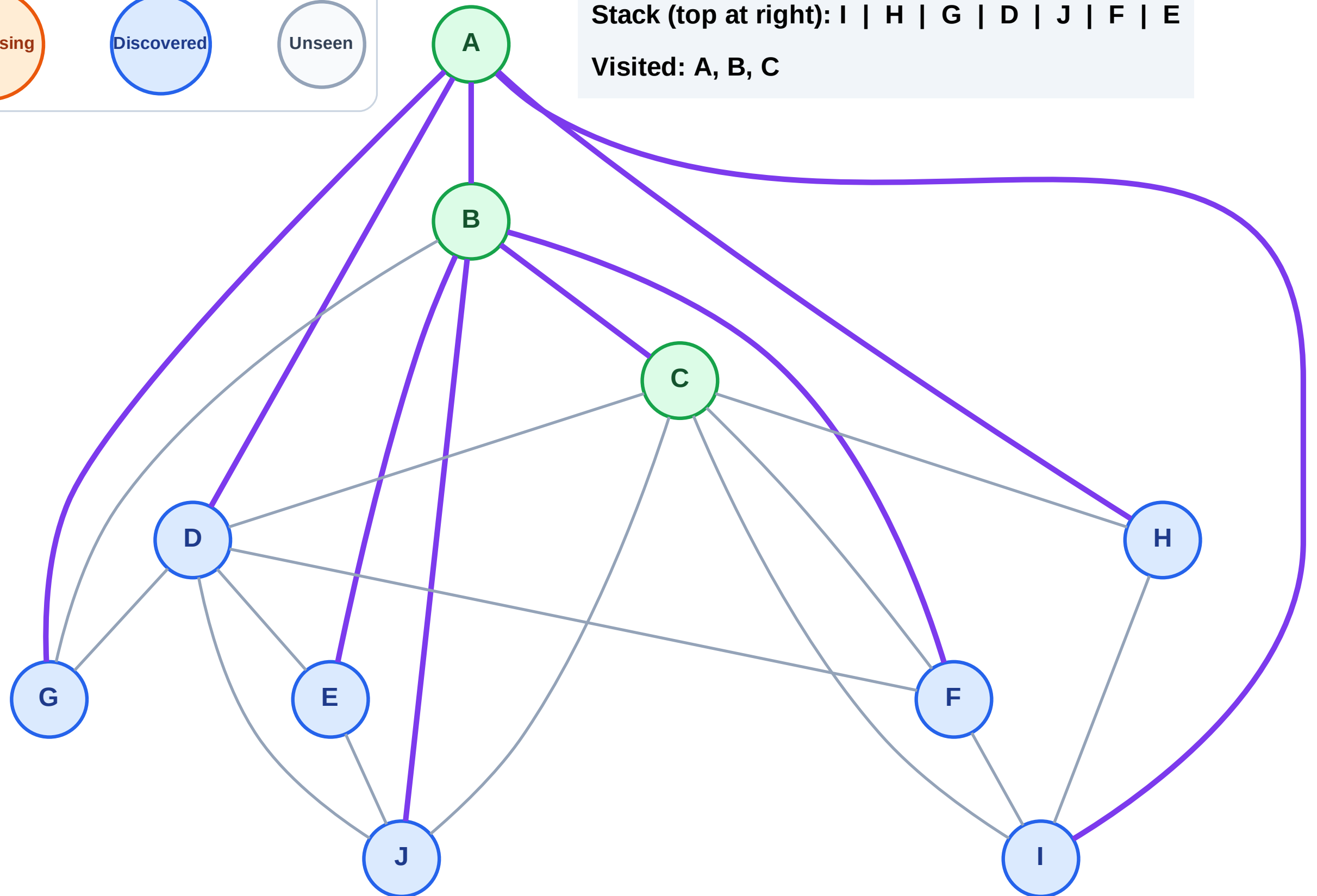
Step 7: No new neighbors from C

Legend



Stack (top at right): I | H | G | D | J | F | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

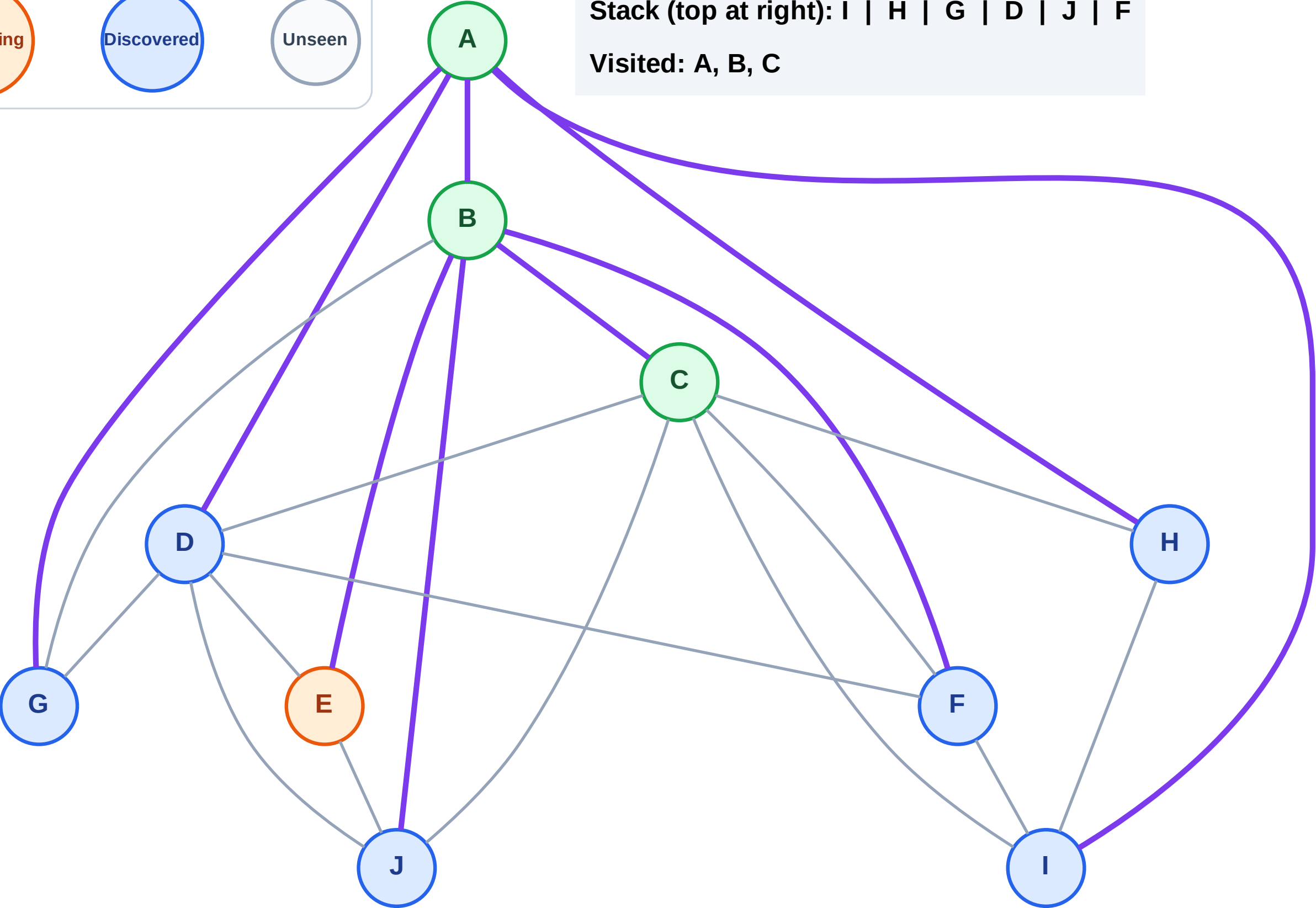
Processing

Discovered

Unseen

Stack (top at right): I | H | G | D | J | F

Visited: A, B, C



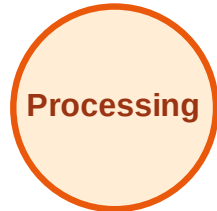
DFS Traversal

Step 9: No new neighbors from E

Legend



Complete



Processing



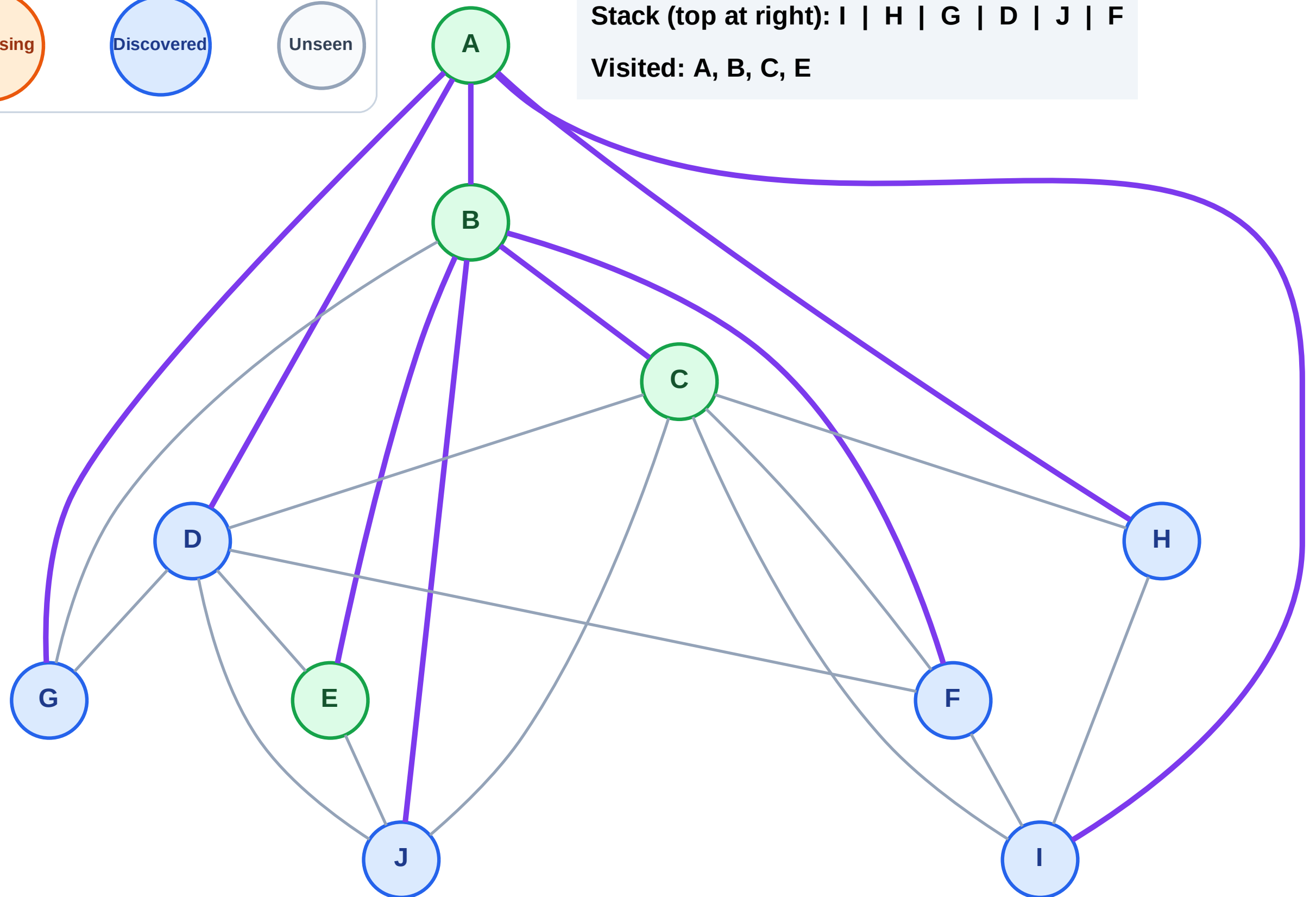
Discovered



Unseen

Stack (top at right): I | H | G | D | J | F

Visited: A, B, C, E



DFS Traversal

Step 10: Process node F

Legend

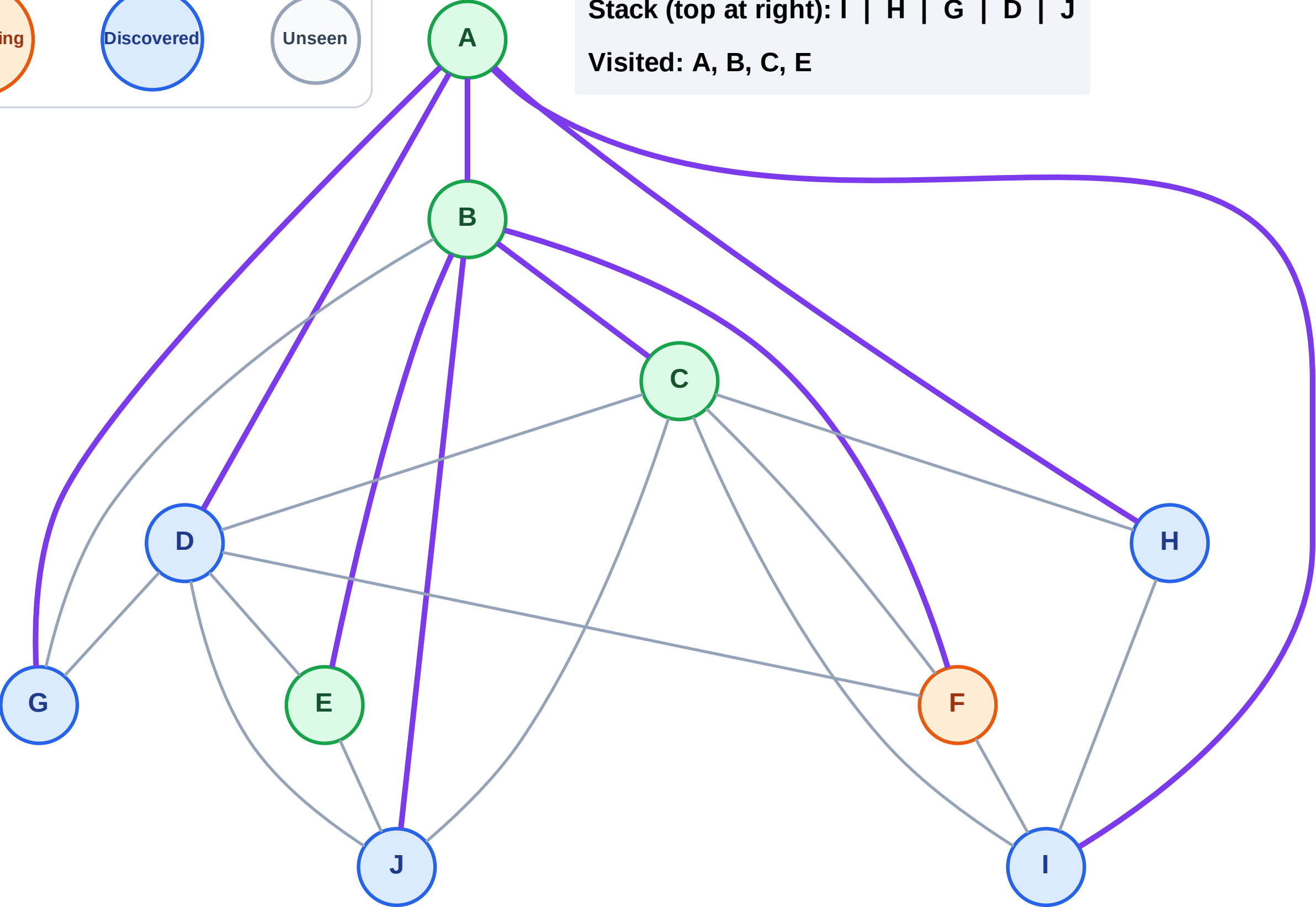
Complete

Processing

Discovered

Unseen

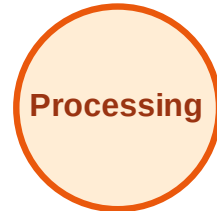
Stack (top at right): I | H | G | D | J
Visited: A, B, C, E



DFS Traversal

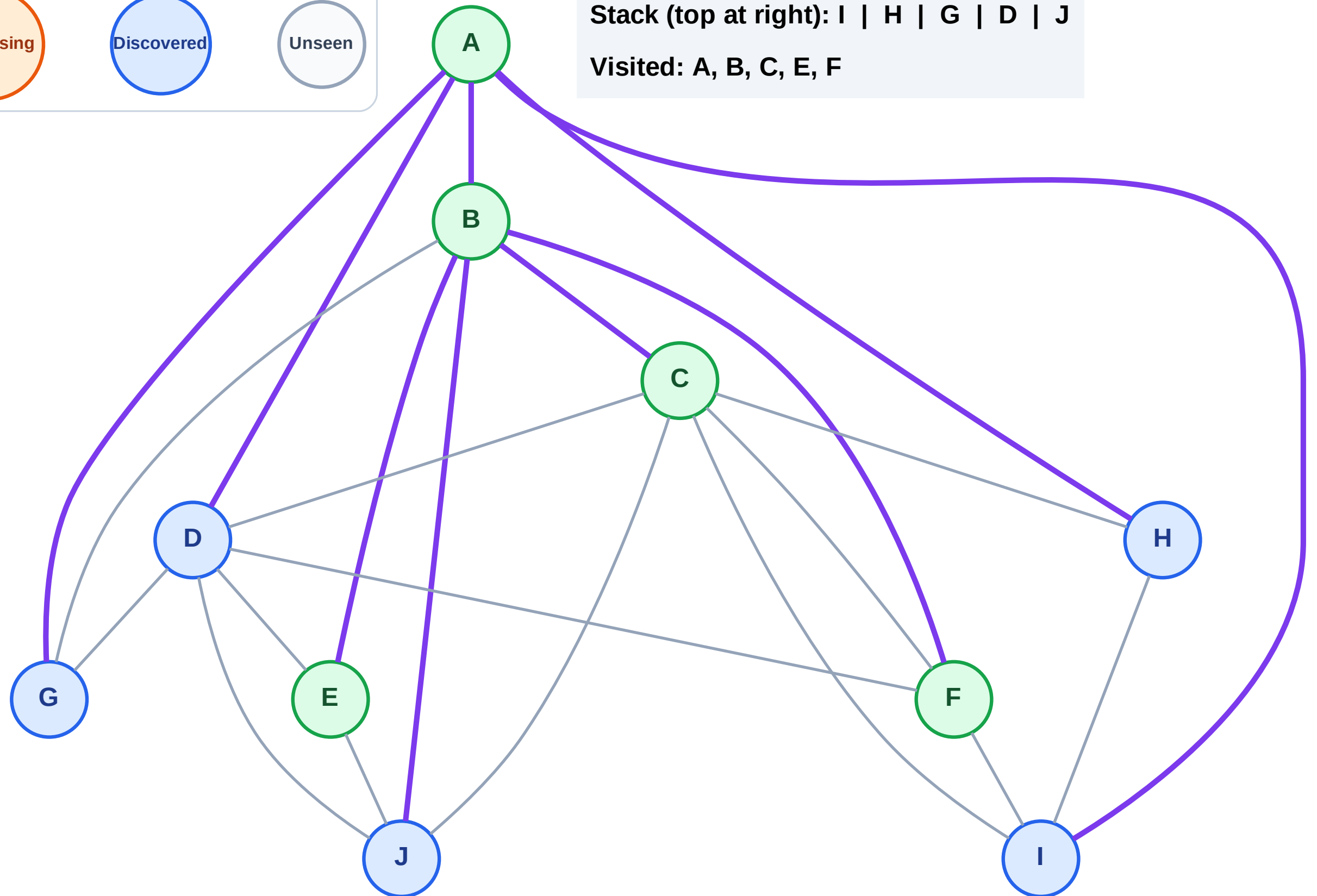
Step 11: No new neighbors from F

Legend



Stack (top at right): I | H | G | D | J

Visited: A, B, C, E, F



DFS Traversal

Step 12: Process node J

Legend

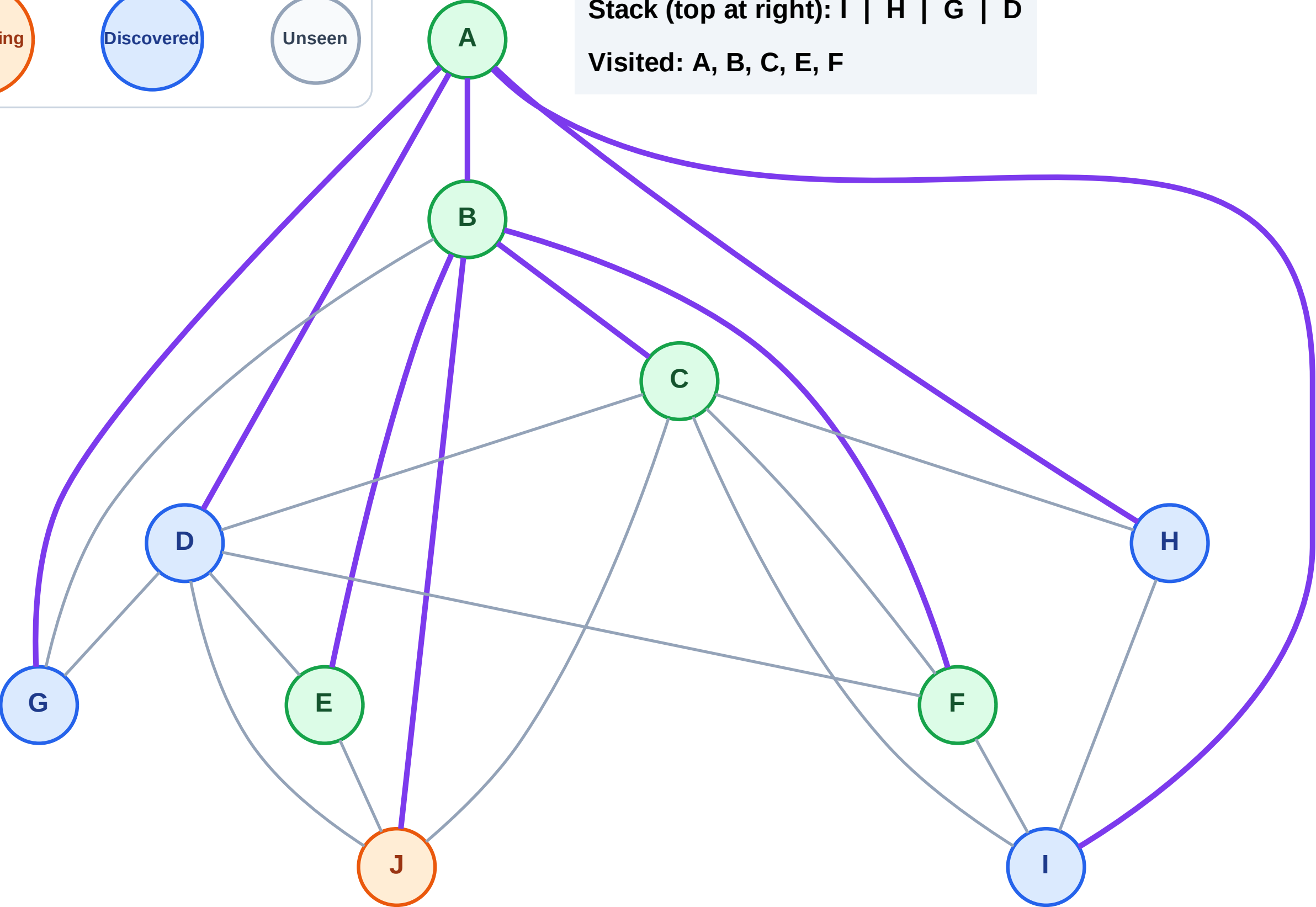
Complete

Processing

Discovered

Unseen

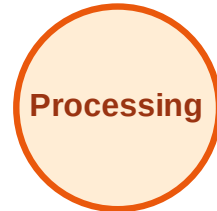
Stack (top at right): I | H | G | D
Visited: A, B, C, E, F



DFS Traversal

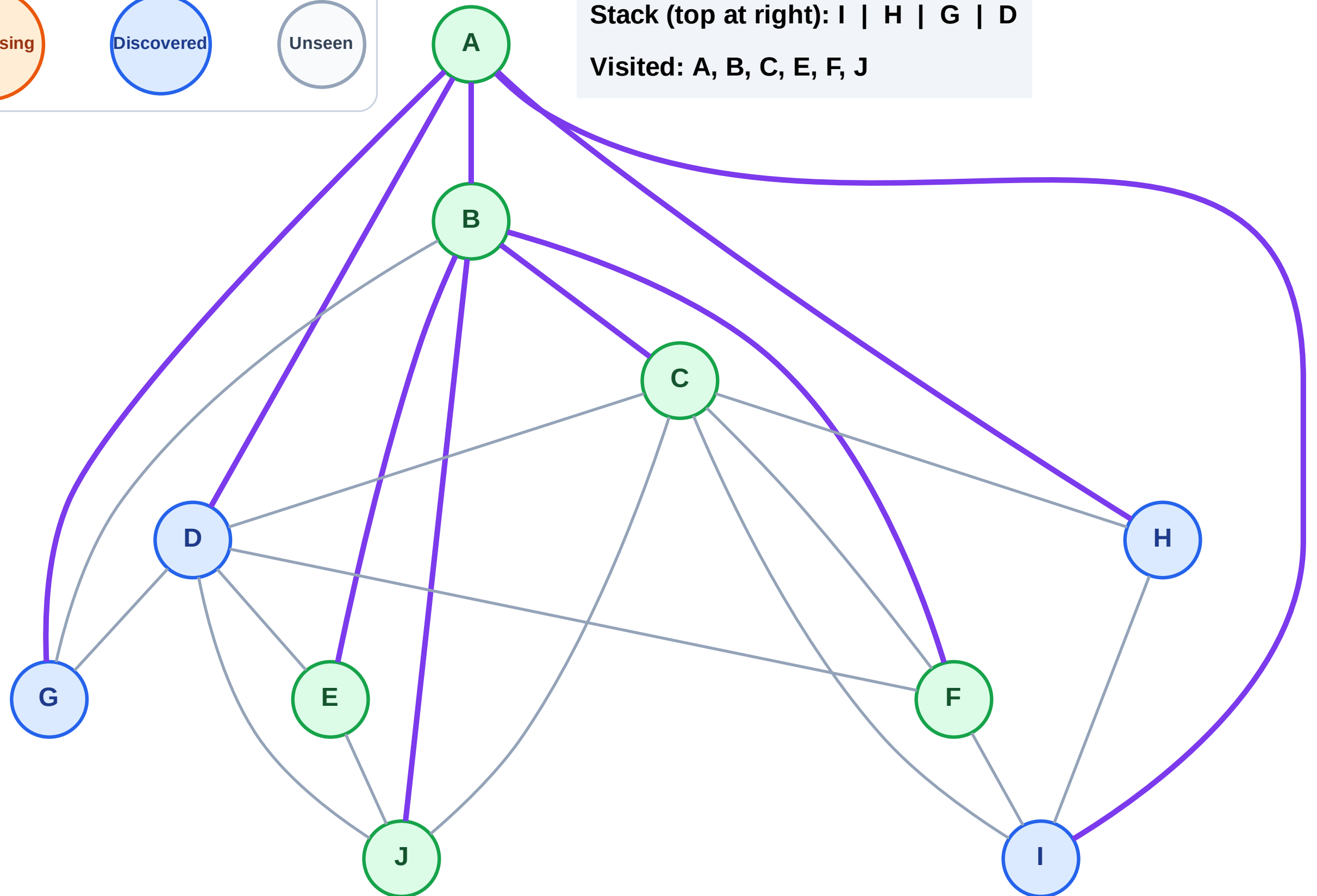
Step 13: No new neighbors from J

Legend



Stack (top at right): I | H | G | D

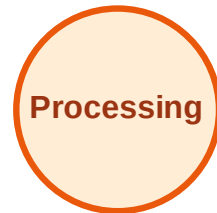
Visited: A, B, C, E, F, J



DFS Traversal

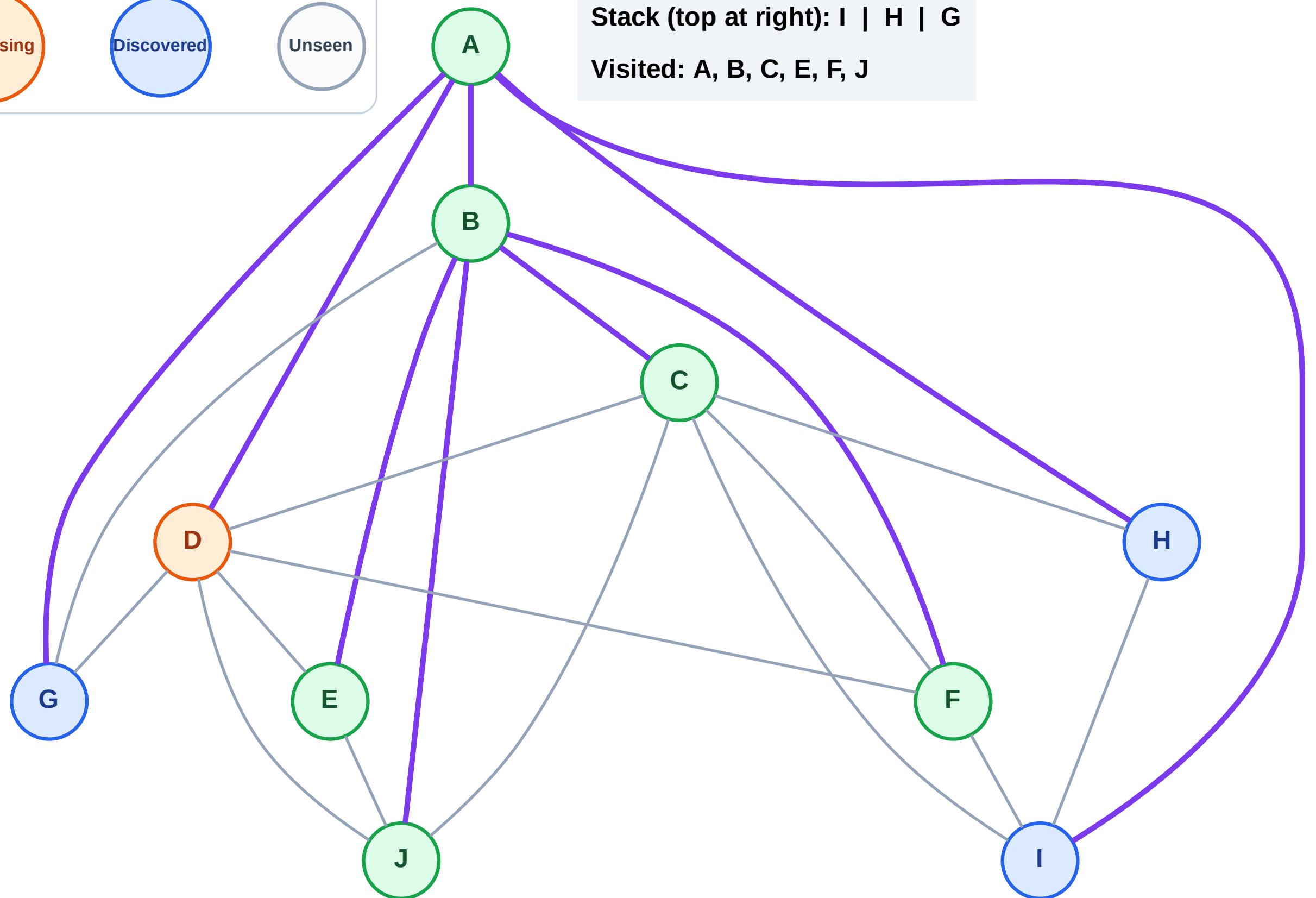
Step 14: Process node D

Legend



Stack (top at right): I | H | G

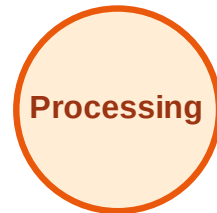
Visited: A, B, C, E, F, J



DFS Traversal

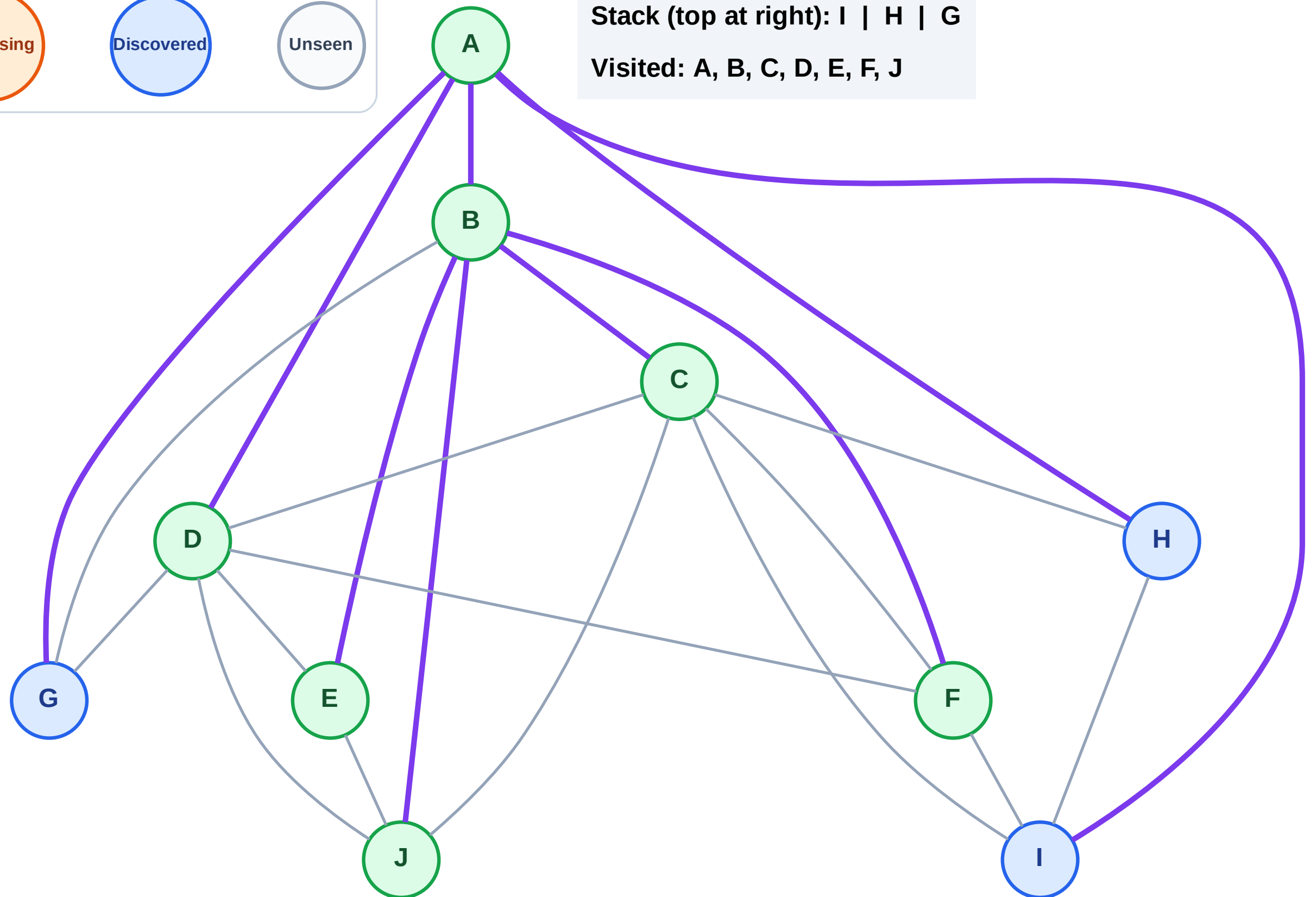
Step 15: No new neighbors from D

Legend



Stack (top at right): I | H | G

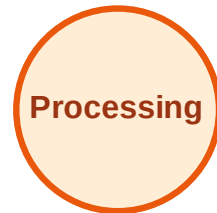
Visited: A, B, C, D, E, F, J



DFS Traversal

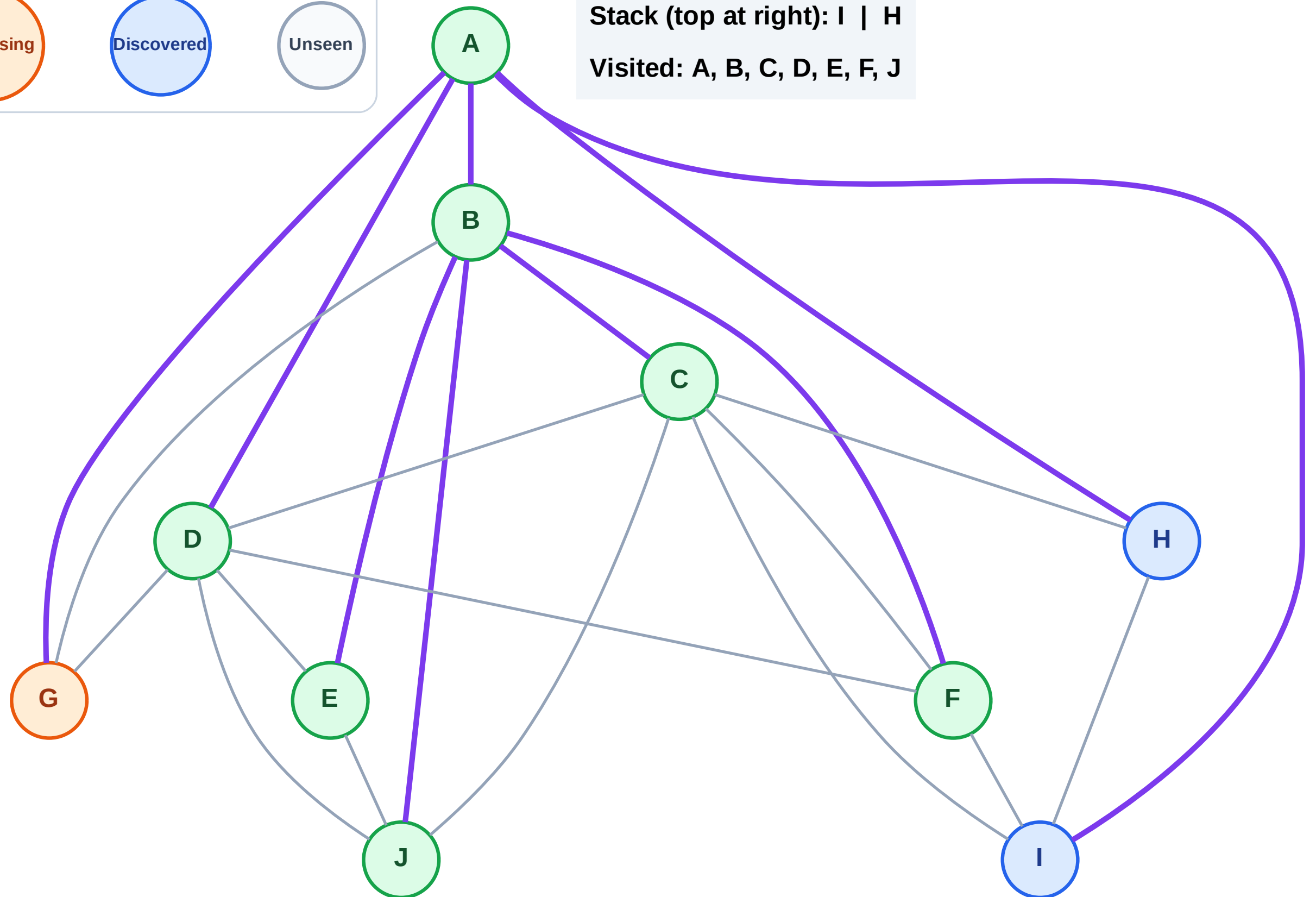
Step 16: Process node G

Legend



Stack (top at right): I | H

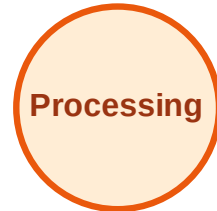
Visited: A, B, C, D, E, F, J



DFS Traversal

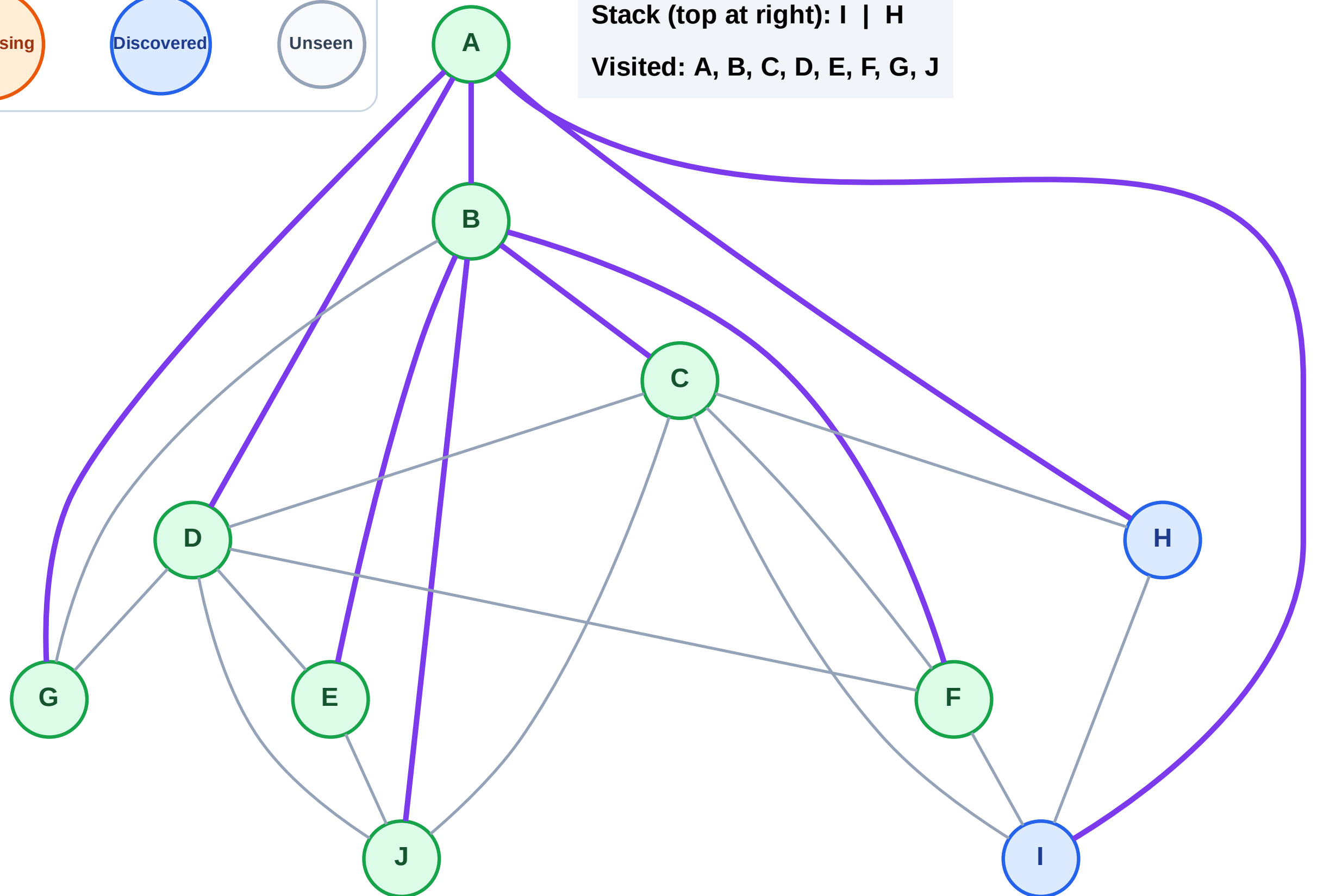
Step 17: No new neighbors from G

Legend



Stack (top at right): I | H

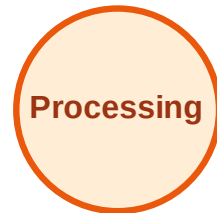
Visited: A, B, C, D, E, F, G, J



DFS Traversal

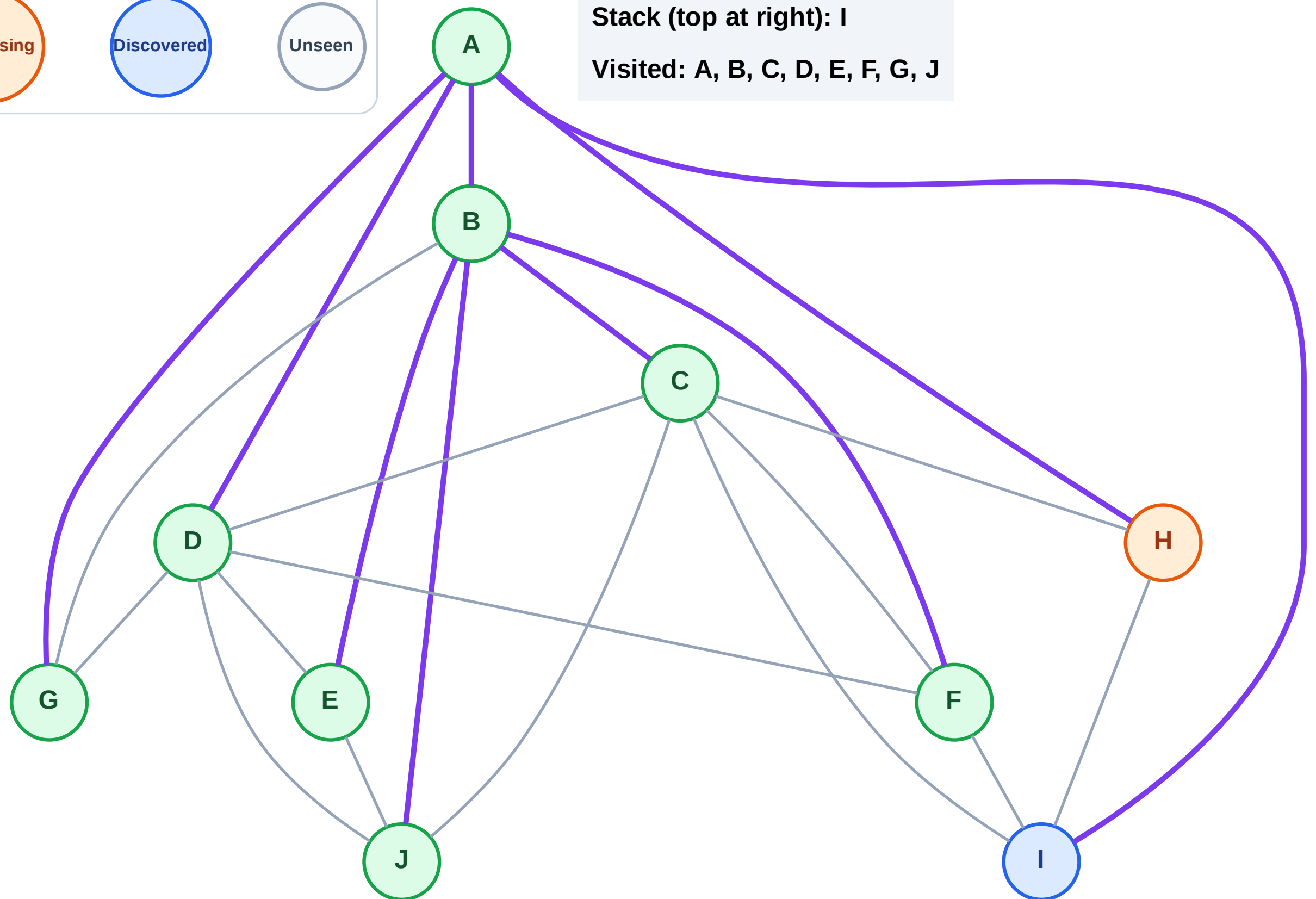
Step 18: Process node H

Legend



Stack (top at right): I

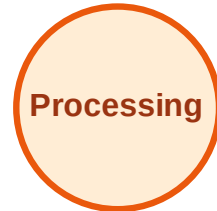
Visited: A, B, C, D, E, F, G, J



DFS Traversal

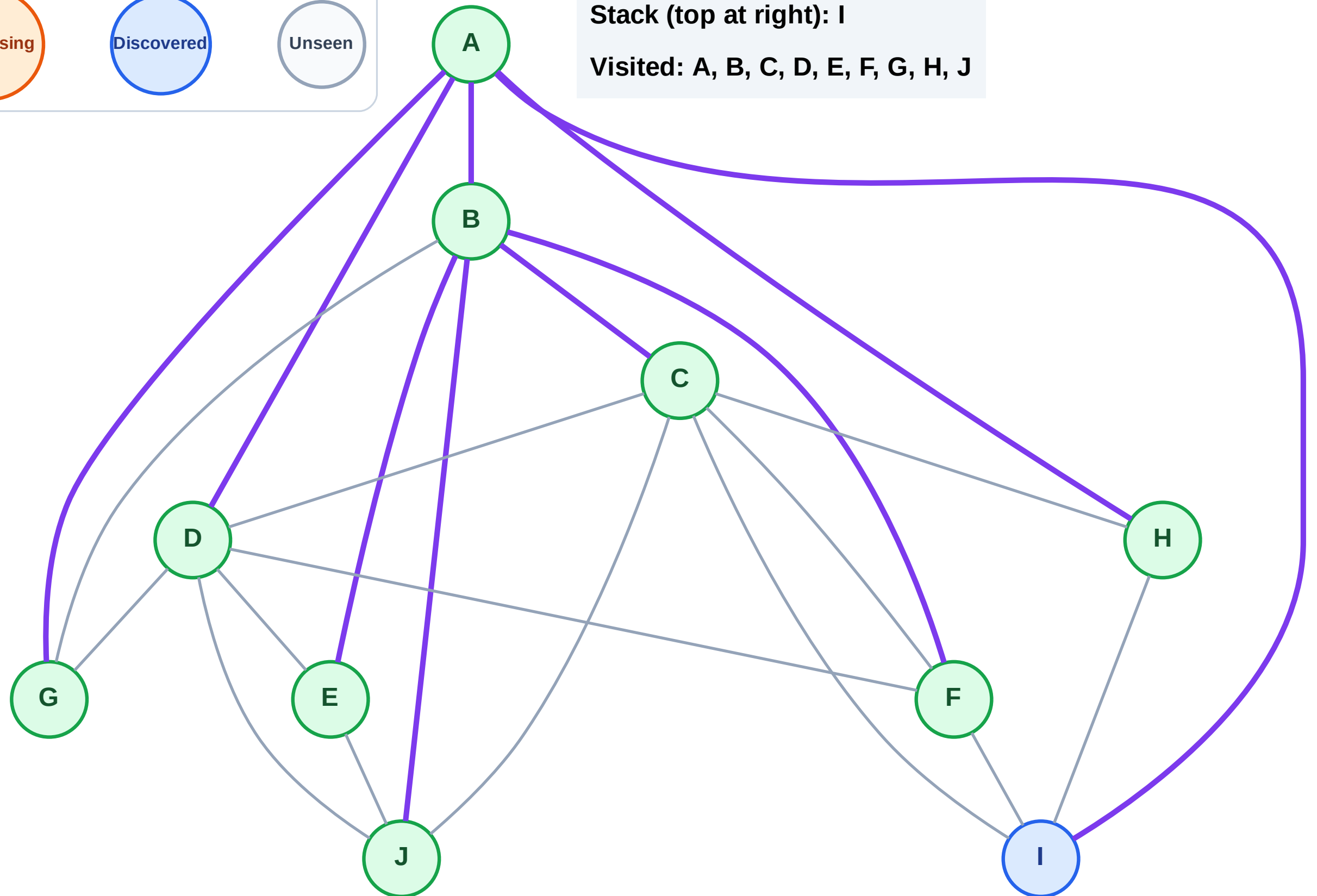
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



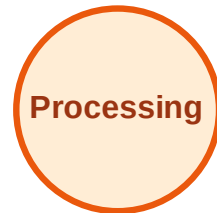
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



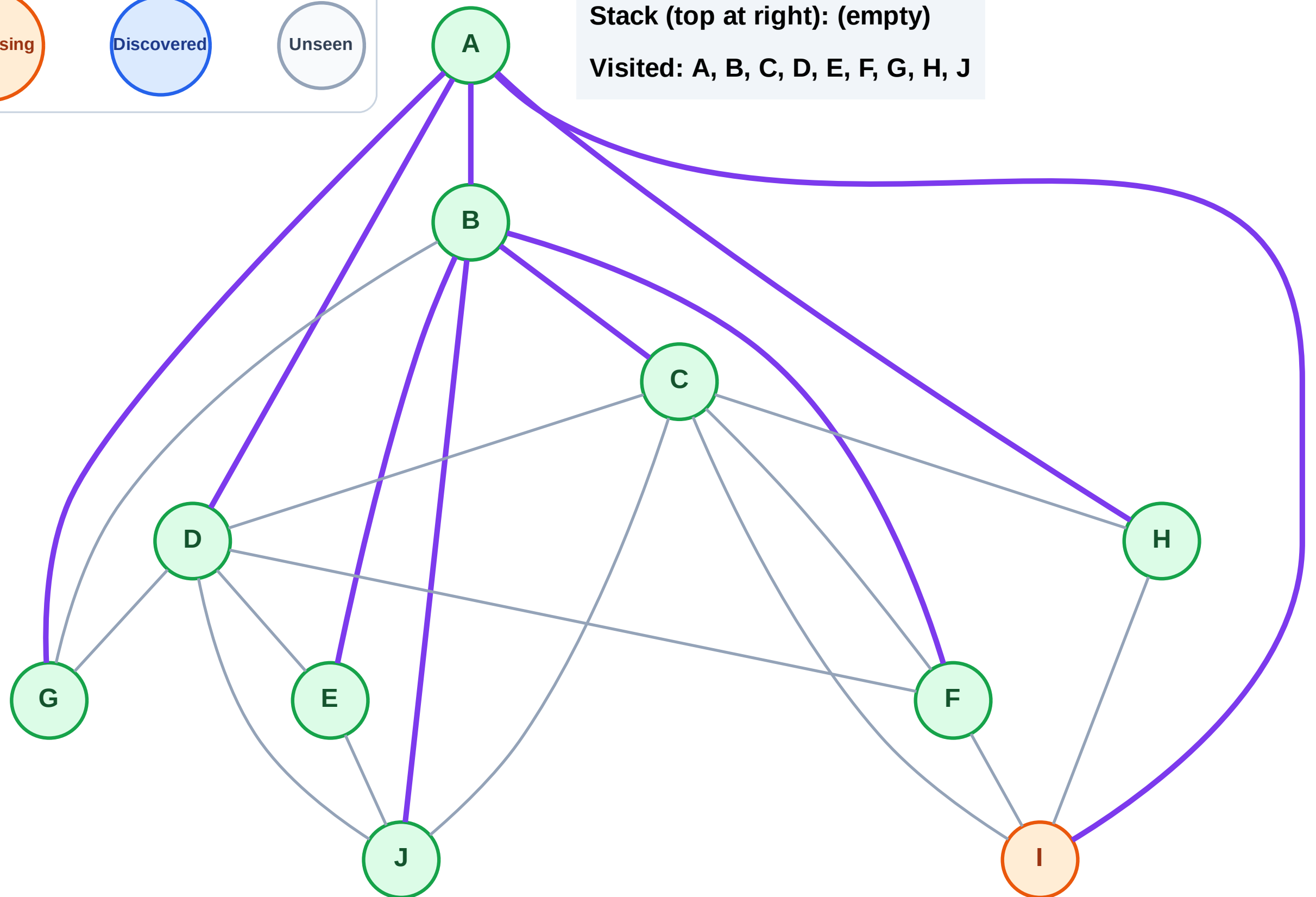
Discovered



Unseen

Stack (top at right): (empty)

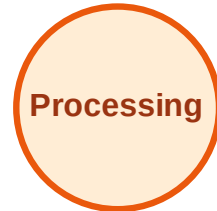
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

